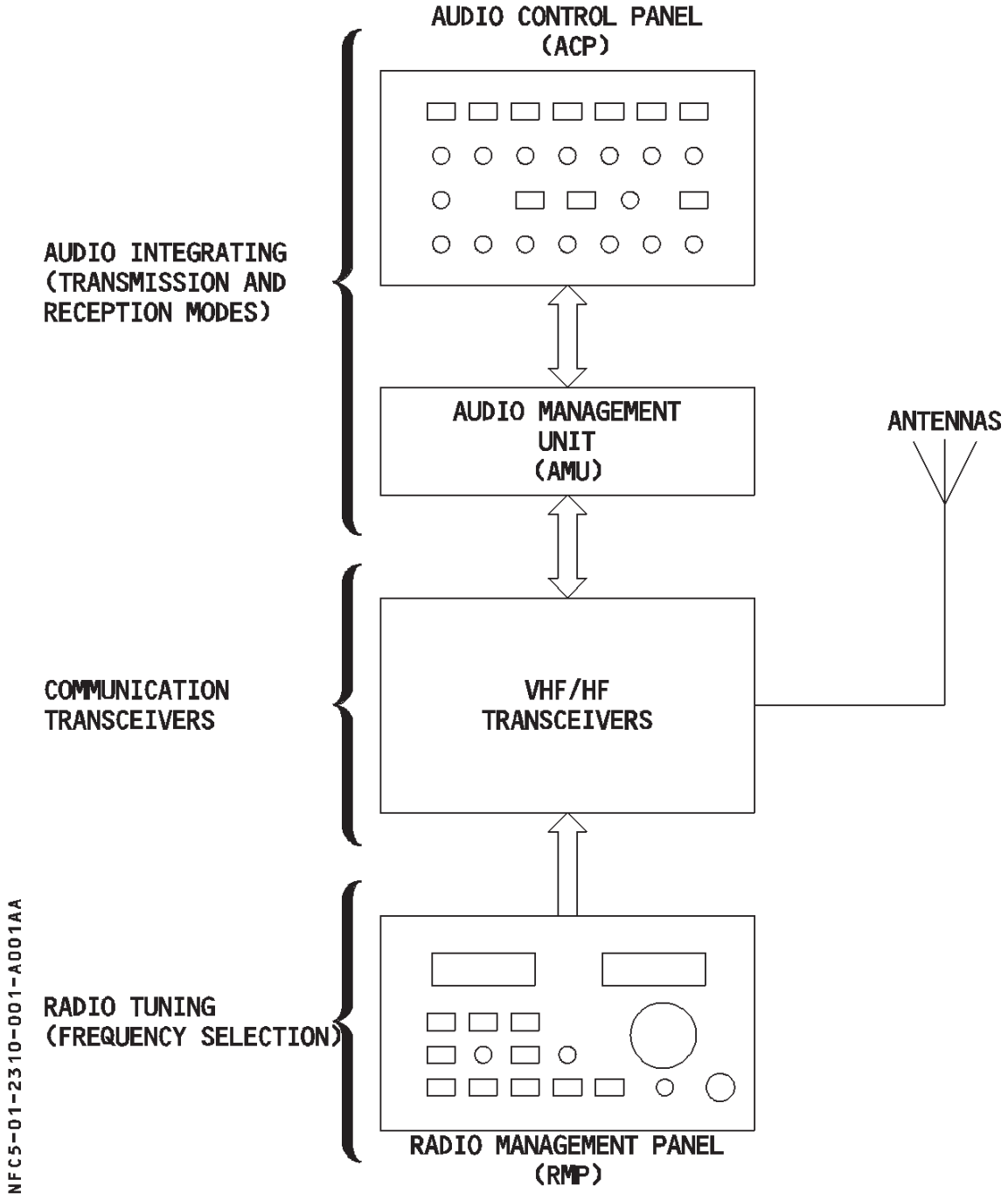
SMARTCOCKPIT.COM

A319/A320/A321

Communications

GENERAL

The communications system comprises the following subsystems :
- VHF/HF tranceivers
- Radio tuning systems (Radio Management Panels).
- Audio integrating system (Audio Management Unit, Audio Control Panels).



VHF/HF/SELCAL

Either of the two Radio Management Panels (RMPs) (third RMP ◁) can be used to tune each transceiver.
To transmit, the flight crew uses the Audio Control Panel (ACP) to select a VHF or HF system. The ACP works through the Audio Management Unit (AMU). Each system is connected to the RMPs, for frequency selection, and to the AMU for connection to the audio integrating and SELCAL (selective calling) systems.

– **VHF**
Two identical VHF communication systems (third VHF system ◁) are installed.
Each system has a transceiver in the avionics compartment, and an antenna on the
R fuselage. Only VHF1 functions in EMER ELEC CONFIG. Its range is from 118.0 to 136.975
R MHZ.
VHF has an alarm which indicates if the microphone is stuck (◁). If a microphone is in the emission position for more than 30 seconds, an interrupted tone sounds for 5 seconds, and the emission is turned off. To reactivate the emission, the crew releases the push-to-talk button and presses it again.

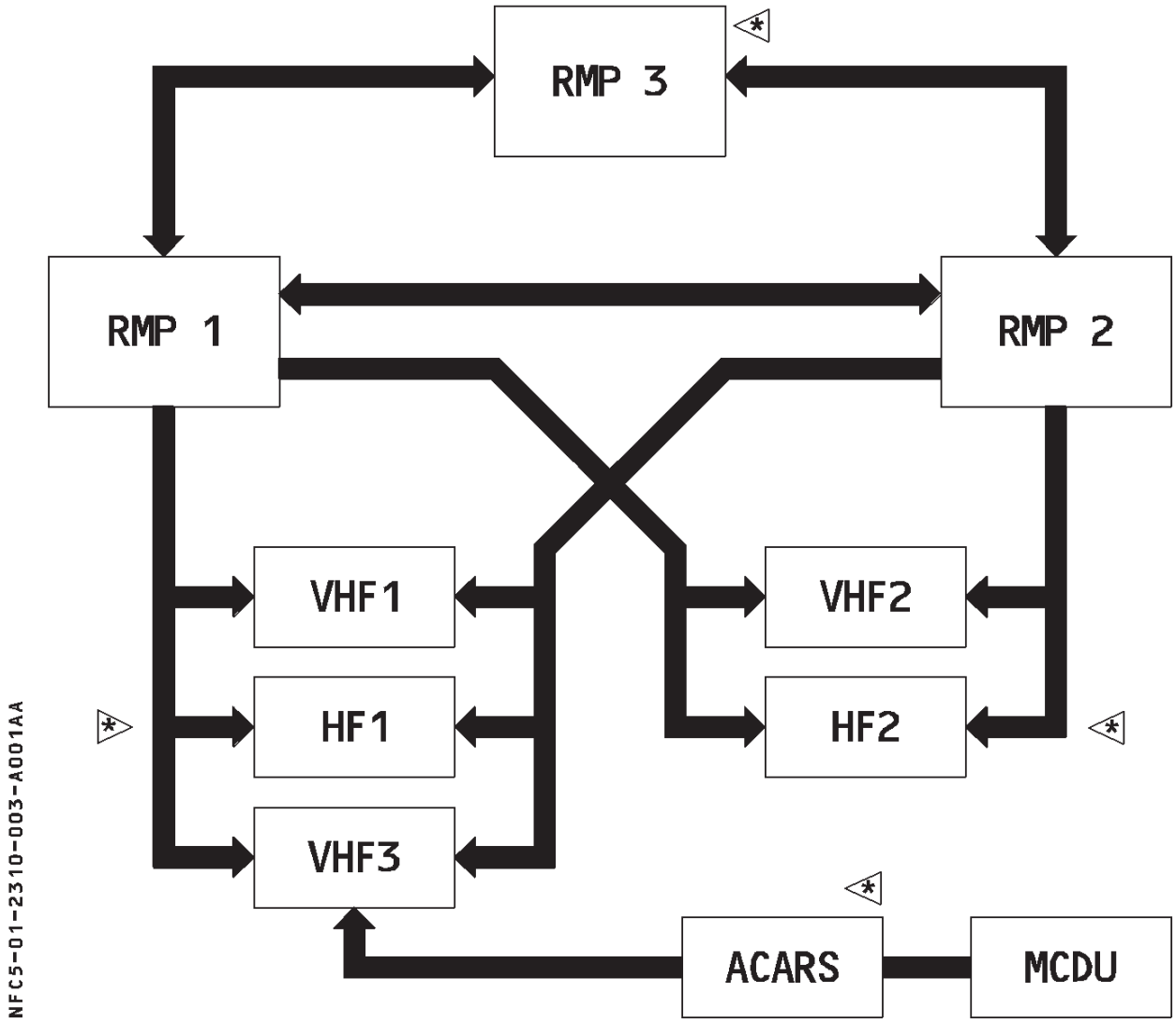
– **HF (◁)**
Two identical HF communication systems are optional. Each has a transceiver in the
R avionics compartment, and a common tuner and antenna in the vertical stabilizer. Its
R range is from 2.8 to 24 MHZ.

– **SELCAL (Selective Calling)**
Upon receiving a call code corresponding to that of the aircraft, the SELCAL system aurally and verbally advises the flight crew that a ground station is calling the aircraft. The aural warning is inhibited during takeoff and landing.

RADIO TUNING

DESCRIPTION

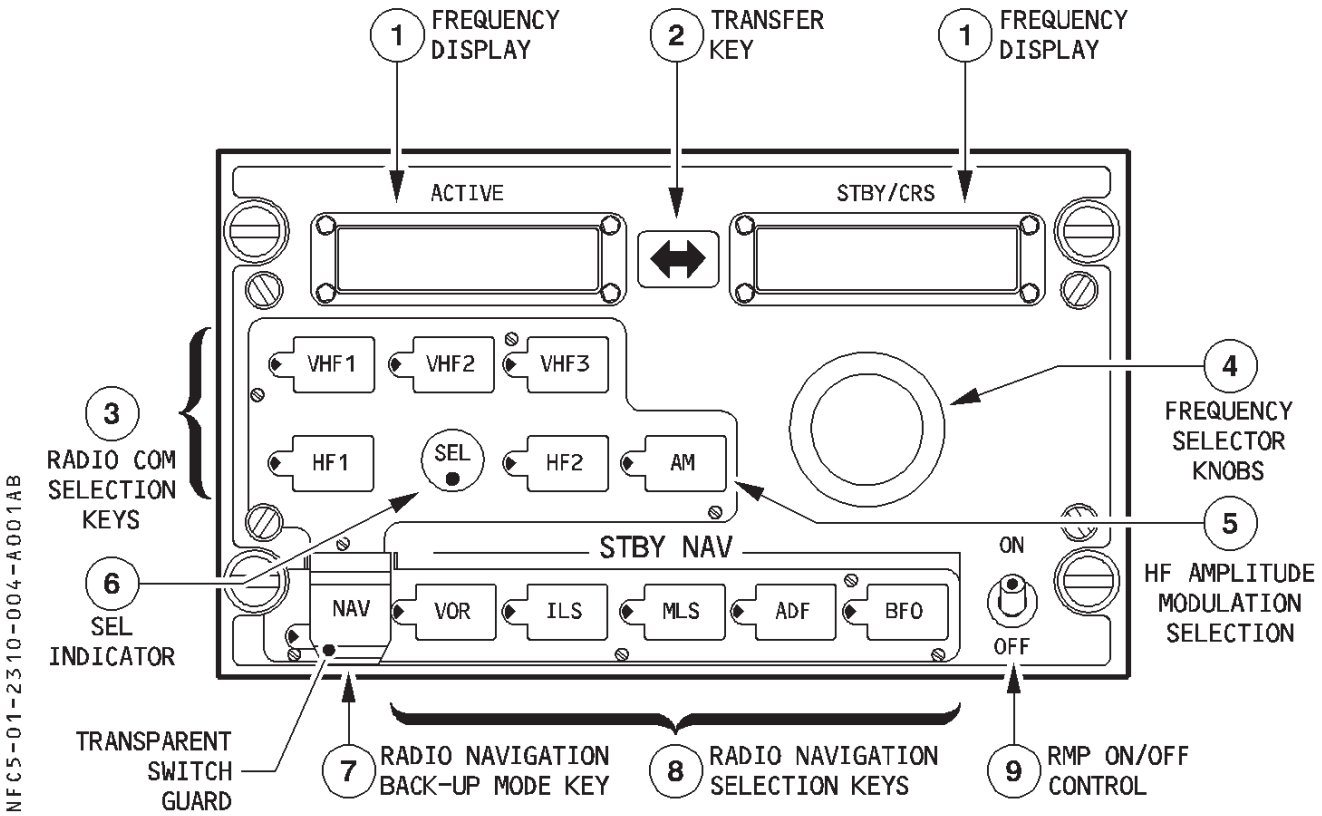
Identical RMPs (Radio Management Panels) :
· Give the flight crew control of all VHF radio communication systems (HF systems ◁).
· Back up to FMGCs for controlling radio navigation systems (Refer to 1.34).
Two RMPs are on the center pedestal (and the third is on the overhead panel ◁).
Each RMP can control any VHF (HF ◁) transceiver. RMP1 and RMP2 are connected directly to all VHF (HF ◁) transceivers, (whereas RMP3 is connected to them via RMP1 and RMP2 ◁). RMPs are connected together so that each RMP is updated to the selections made on other RMPs.
Only RMP1 functions in EMER ELEC CONFIG.



R If one RMP fails, the remaining one controls all the VHF (HF ◁) tranceivers.

RADIO MANAGEMENT PANEL

R



(1) Frequency displays

The ACTIVE display window shows the active frequency of the selected radio, which is identified by a green light on the selection key.
The STBY/CRS (standby/course) display window shows a standby frequency that the pilot can activate by pressing the transfer key or change by rotating the tuning knobs. (For a description of the CRS function (see 1.34)).

(2) Transfer key

Pressing this key moves the active frequency to the standby window and the standby frequency to the active window.
This tunes the selected receiver to the new active frequency.

(3) Radio com selection keys

When the pilot presses one of these keys :
· The ACTIVE window displays the frequency set on that radio.
· The STBY/CRS window displays the selected standby frequency or course.
· The selected key displays a green monitor light.

④ Frequency selector knobs

The pilot uses these concentric knobs to select the STBY frequency or CRS.
The outer knob controls whole numbers ; the inner knob controls decimal fractions.

⑤ AM pb sw

If the aircraft has HF radios and the flight crew has selected an HF tranceiver, this switch selects the AM mode. (The default mode is the SSB, or single side-band, mode).
This key displays a green monitor light when the AM mode is active.

⑥ SEL indicator

The SEL indicator glows white on both RMPs when a transceiver normally associated with one RMP is tuned by another :
· VHF1 (VHF 3, HF 1 ◁) tuned by RMP2,
· VHF2 (HF2 ◁) tuned by RMP1.

⑦ NAV pb sw (with transparent switchguard)

The pilot presses this key to be able to select navigation receivers and courses through the RMP. It does not affect the selection of communication radios and their frequencies. (Refer to 1.34) for additional information.

⑧ Radio navigation selection keys

The pilot presses one of these keys to select a navigation radio to control through this RMP. This turns on the green monitor light in the key.
(Refer to 1.34) for additional information.

⑨ ON/OFF sw

This switch controls the power supply of the RMP.

Note : RMP3 is able to control VHF and HF transceivers through RMP1 and RMP2 even when either of the latter is OFF.

④ Frequency selector knobs

The pilot uses these concentric knobs to select the STBY frequency or CRS.
The outer knob controls whole numbers ; the inner knob controls decimal fractions.

⑤ AM pb sw

If the aircraft has HF radios and the flight crew has selected an HF tranceiver, this switch selects the AM mode. (The default mode is the SSB, or single side-band, mode).
This key displays a green monitor light when the AM mode is active.

⑥ SEL indicator

The SEL indicator glows white on both RMPs when a transceiver normally associated with one RMP is tuned by another :
· VHF1 tuned by RMP2 or RMP3,
· VHF2 tuned by RMP1 or RMP3.
· VHF3, HF1, HF2 (◁) tuned by RMP1 or RMP2.

⑦ NAV pb sw (with transparent switchguard)

The pilot presses this key to be able to select navigation receivers and courses through the RMP. It does not affect the selection of communication radios and their frequencies. (Refer to 1.34 for additional information).

⑧ Radio navigation selection keys

The pilot presses one of these keys to select a navigation radio to control through this RMP. This turns on the green monitor light in the key.
(Refer to 1.34 for addition information).

⑨ ON/OFF sw

This switch controls the power supply to the RMP.

Note : RMP3 is able to control VHF and HF transceivers through RMP1 and RMP2 even when they are OFF.

GENERAL

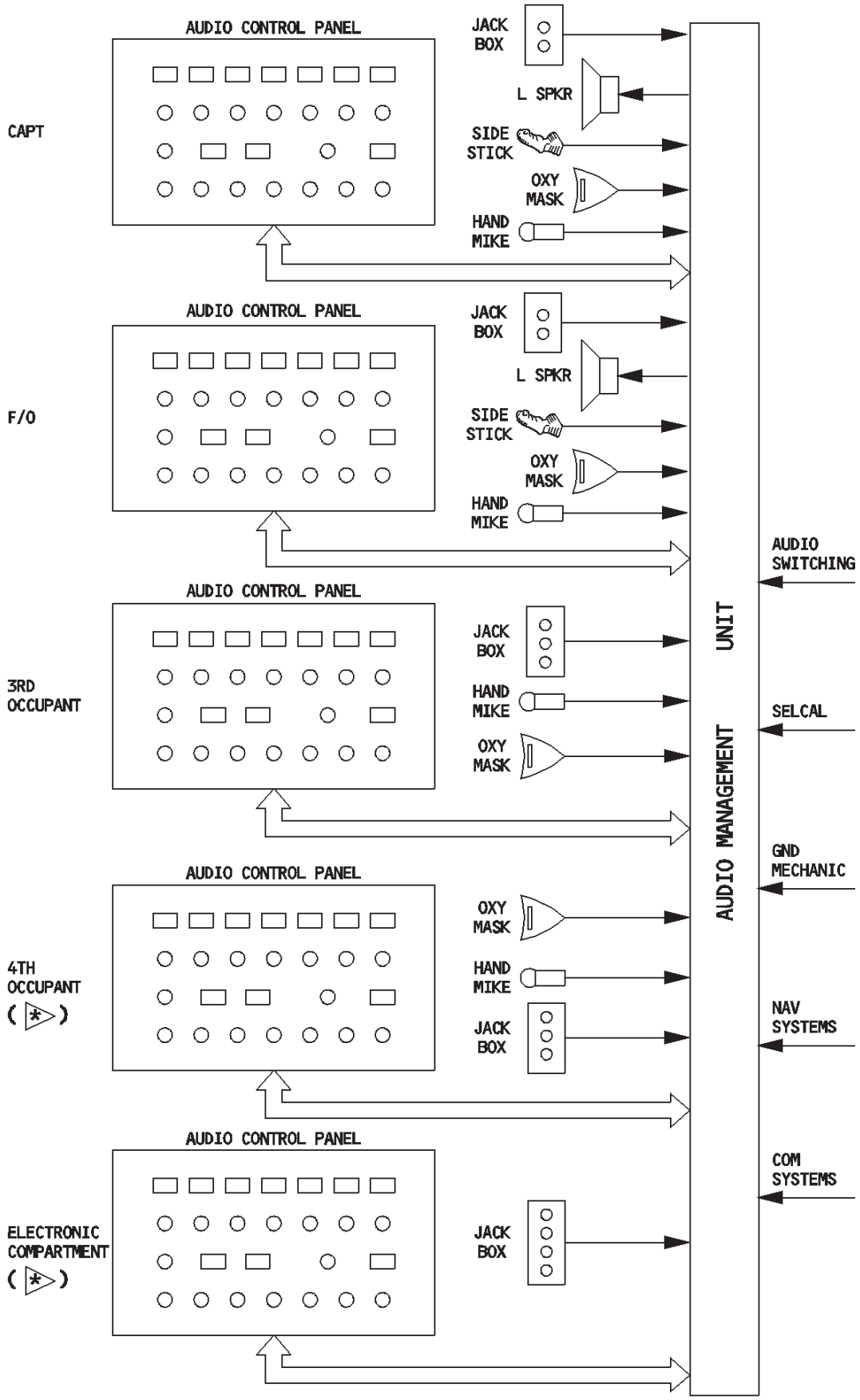
Intercommunications is divided into two main systems:
- the audio management system.
- the cabin intercommunication data system.

AUDIO MANAGEMENT SYSTEM

The audio management system allows the flight crew to use :
- all the radio communication and radio navigation facilities installed on the aircraft in transmission and reception mode.
- the interphone systems
- the call systems
- the passenger address system

The audio management system includes :
- an audio management unit (AMU)
- three audio control panels (ACPs) (fourth and fifth optional)
- sockets at each station
  - headset jack and boomset connector (hand microphone connector (◁) for pilot, copilot, and third occupant
  - headset jack for fourth occupant
- one interphone jack at the ground power receptacle
- boomsets for the pilot, copilot, and third occupant, and three hand microphones (fourth (◁))
- three cockpit oxygen mask microphones
- one radio press-to-talk switch on each sidestick
- one SELCAL code selection panel (avionics compartment)
- two cockpit loudspeakers with separate volume controls
- if installed, a jack panel in the electronic compartment that groups the headset jack, service interphone jack, hand mike connector, and boomset
- an audio switching facility

If audio channel 1 or 2 fails due to a failure either in an ACP or the corresponding AMU, the crew can use the AUDIO SWITCHING selector to select the third audio channel.



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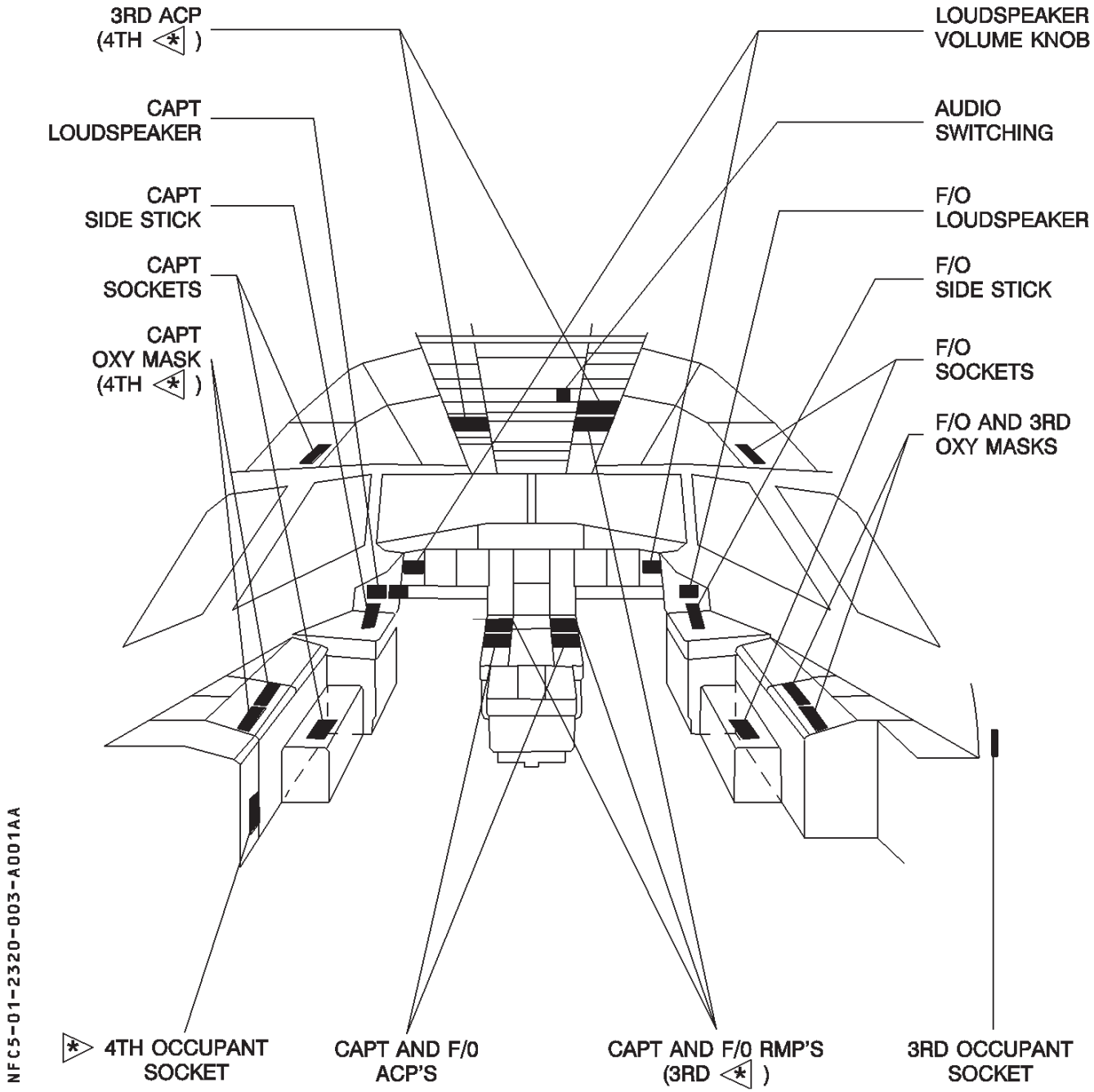
LOCATION OF COMPONENTS (PILOT'S STATION)



CABIN INTERCOMMUNICATION DATA SYSTEM

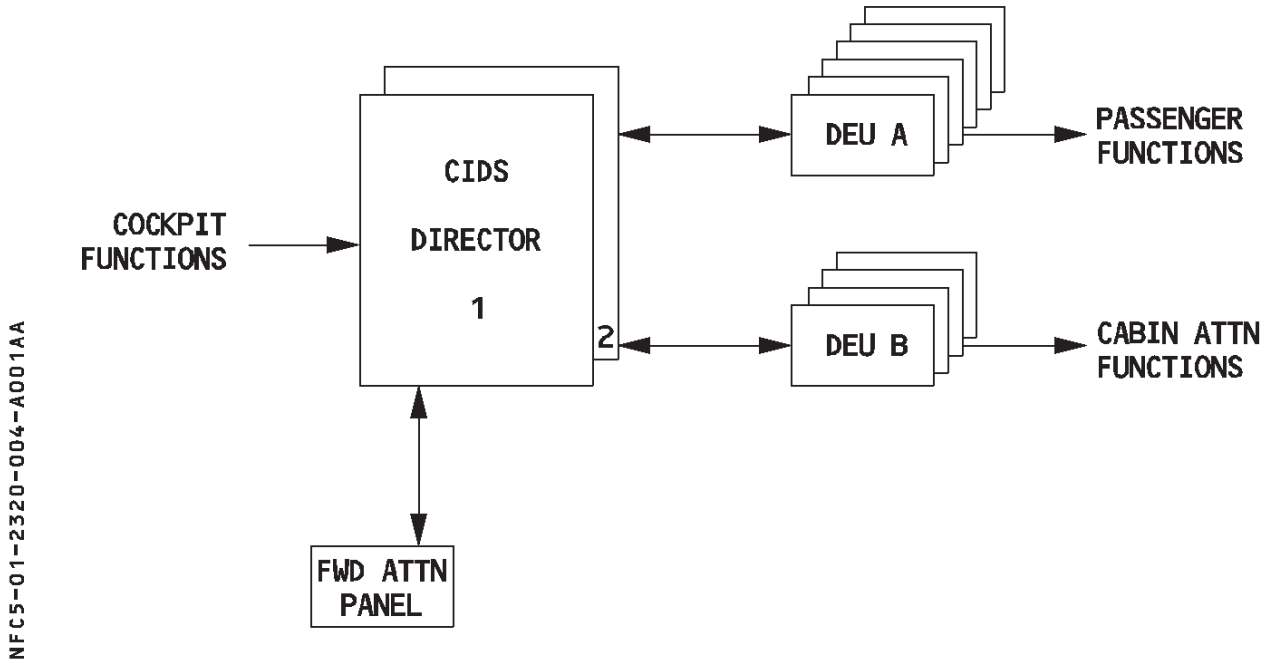
The Cabin Intercommunication Data System (CIDS) transmits, controls, and processes signals for the following cabin systems :

- cabin and service interphone
- passenger address
- passenger lighted sign
- reading light
- general cabin illumination
- emergency evacuation signalling
- lavatory smoke indication
- passenger entertainment music and video ◁
- R escape slide bottle pressure monitoring ◁

The CIDS has the following main components :

- two CIDS directors connected in parallel, one active and the other in standby.
- forward attendant panel for control of the cabin systems.
- programming and test module that allows the system to be reprogrammed after changes are made in the cabin configuration.

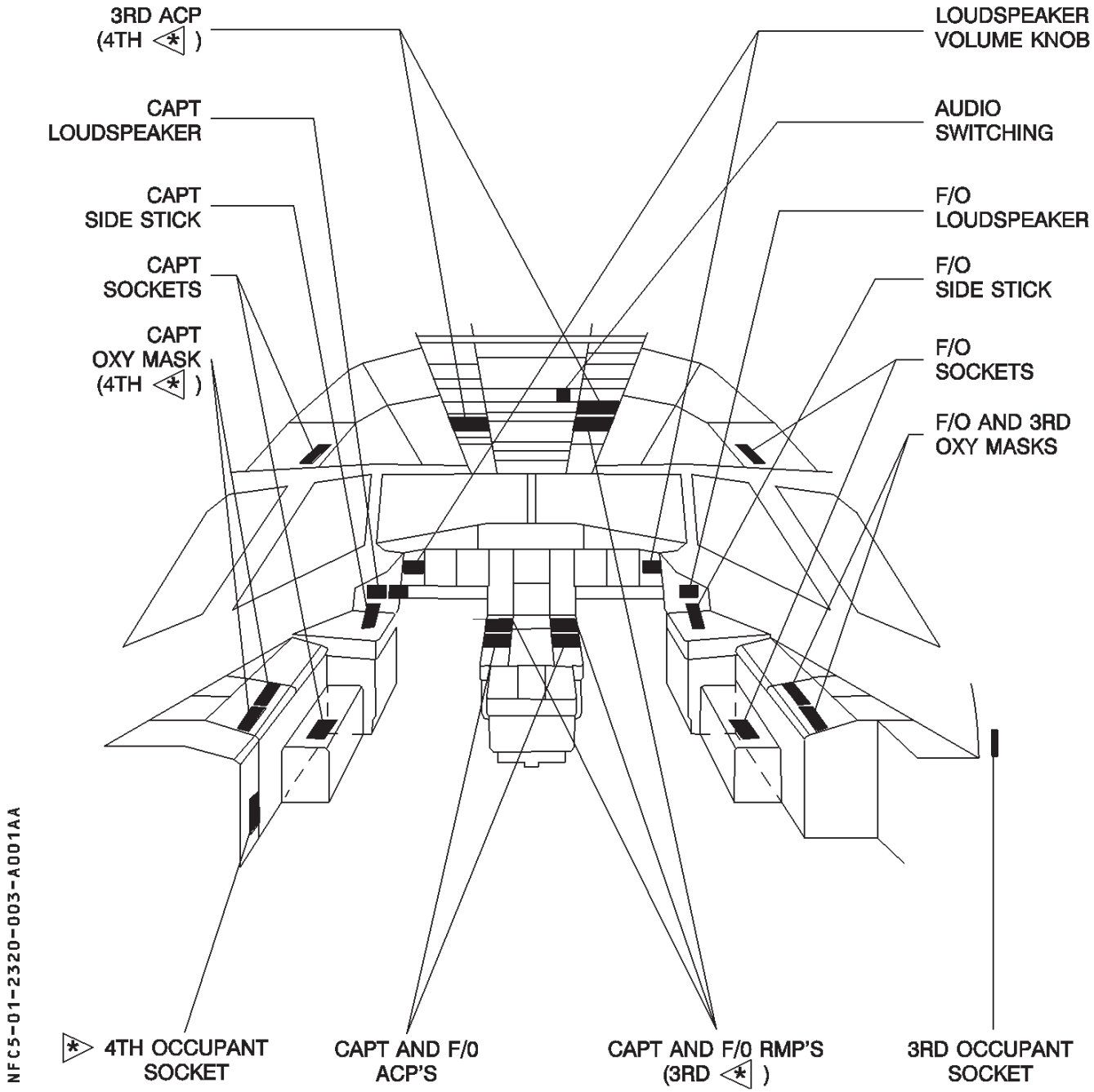
FOR INFO



Decoder/Encoder Units (DEUs) are linked to the two directors.

- *Type A units (for passengers) are along each side of the passenger cabin.*
  The loudspeakers, lighted signs, call buttons, call lights and general illumination ballast units are divided into small groups each connected to a type A DEU.
- *Type B units (for attendants) are near the exit doors. The area call panels, attendant handsets, slide and door pressure sensors, and attendant indicator panels are connected to type B DEUs.*

LOCATION OF COMPONENTS (PILOT'S STATION)



CABIN INTERCOMMUNICATION DATA SYSTEM

The Cabin Intercommunication Data System (CIDS) transmits, controls, and processes signals for the following cabin systems :

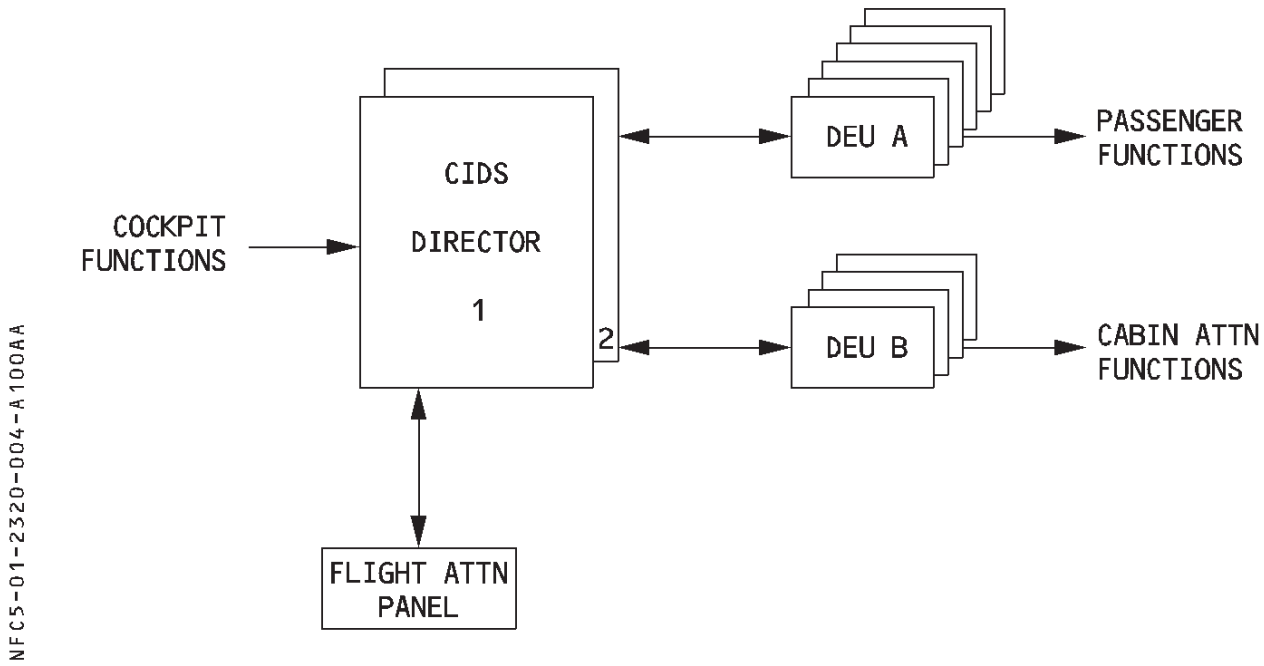
- Cabin and service interphone
- Passenger address
- Passenger lighted signs
- Reading lights (LED technology)
- General cabin illumination
- Emergency evacuation signalling
- Lavatory smoke detectors and indicators
- Passenger entertainment music and video
- Escape slide bottle pressure monitoring
- R Vacuum system control function
- R Cargo and lavatory smoke detection function

The CIDS has the following main components :

- Two CIDS directors connected in parallel ; one active, and the other on standby.
- R Flight Attendant Panel (FAP) to monitor and control the cabin systems.
- Attendant Indication Panels (AIP), After Attendant Panels (AAP) and Area Call Panels (ACP)
- Decoder/Encoder Units (DEUs) linked to the two directors.
  - Type A units (for passengers) installed along the cabin.
    The loudspeakers, lighted signs, call buttons, call lights and general illumination ballast units are divided into small groups, each connected to a type A DEU.
  - R Type B units (for attendants) installed near the exit doors. The Area Call panels, smoke detectors, attendant handsets, slide and door pressure sensors, and attendant indicator panels, are connected to type B DEUs.

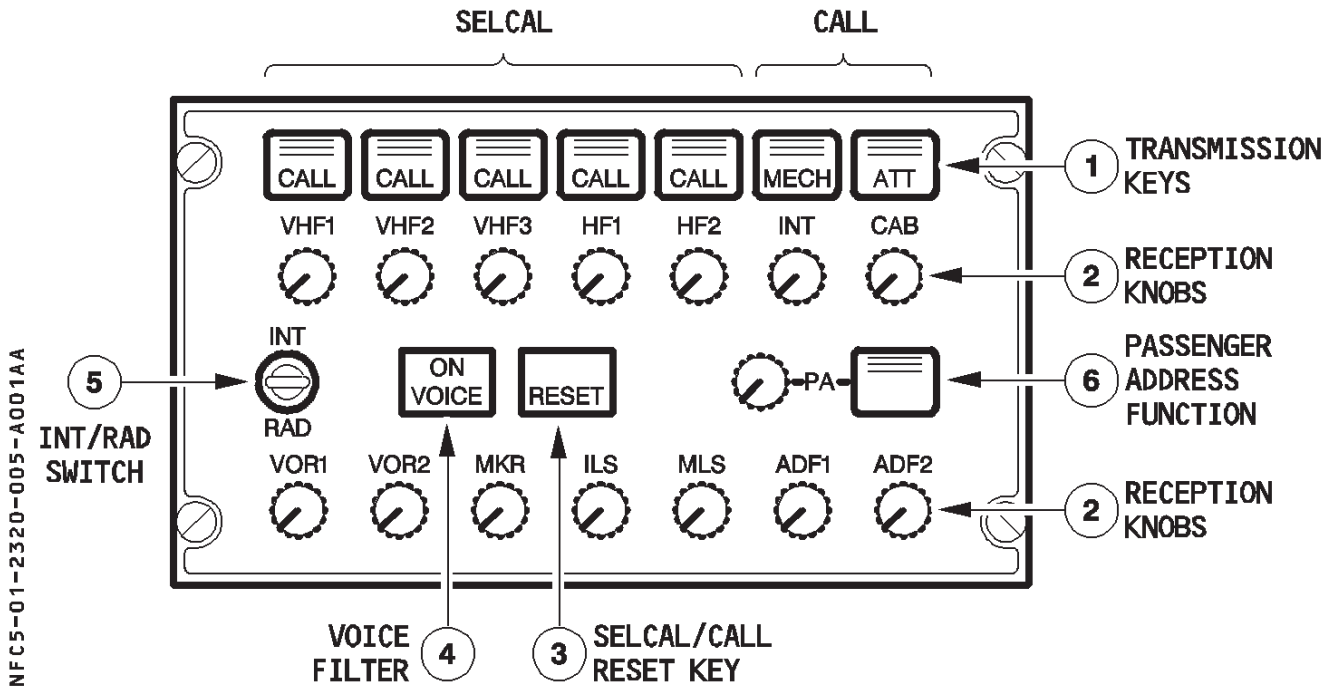
FOR INFO

R



CONTROLS

AUDIO CONTROL PANEL



(1) Transmission keys

Pressed	:	The associated channel is selected for transmission. The three green lines come on. The pilot deselects the channel by pressing the pushbutton again, or by selecting another channel.
CALL	:	The legend flashes amber (and buzzer sounds) when the SELCAL system detects a call.
MECH	:	The legend flashes amber (and buzzer sounds) for a call from the nose gear bay. The MECH light goes off after 60 seconds, if it is not reset.
ATT	:	The legend flashes amber (and buzzer sounds) for a call from a cabin attendant. The ATT light goes off after 60 seconds, if it is not reset.

② Reception knobs

- Pressing and releasing the knob (knob out) selects the associated audio reception channel and the integral white light comes on.
- Rotating the knob adjusts the volume.
- R The INTEG LT knob controls the brightness.
- Pressing the knob (knob stays in) disconnects the associated audio reception channel.

R *Note : To receive DME audio navigation signals that are associated with an ILS or MLS*
R *station : The flight crew must not only select the ILS (or LS) pushbutton on the*
R *FCU, but also the VOR reception channel on the ACP. It is possible that the audio*
R *signals of the VOR, the ILS, and the DME (associated with ILS) could all sound*
R *at the same time.*

③ SELCAL/CALL RESET key

Pressing this key extinguishes CALL, MECH, and ATT lights.

④ ON VOICE key

This key allows the flight crew to inhibit the audio navigation signals (VOR, ADF)
Pressing this key filters out ident signals and turns on the green ON light.

⑤ INT/RAD sw

This switch operates as a press-to-talk switch for boom mike or oxygen mask mike.

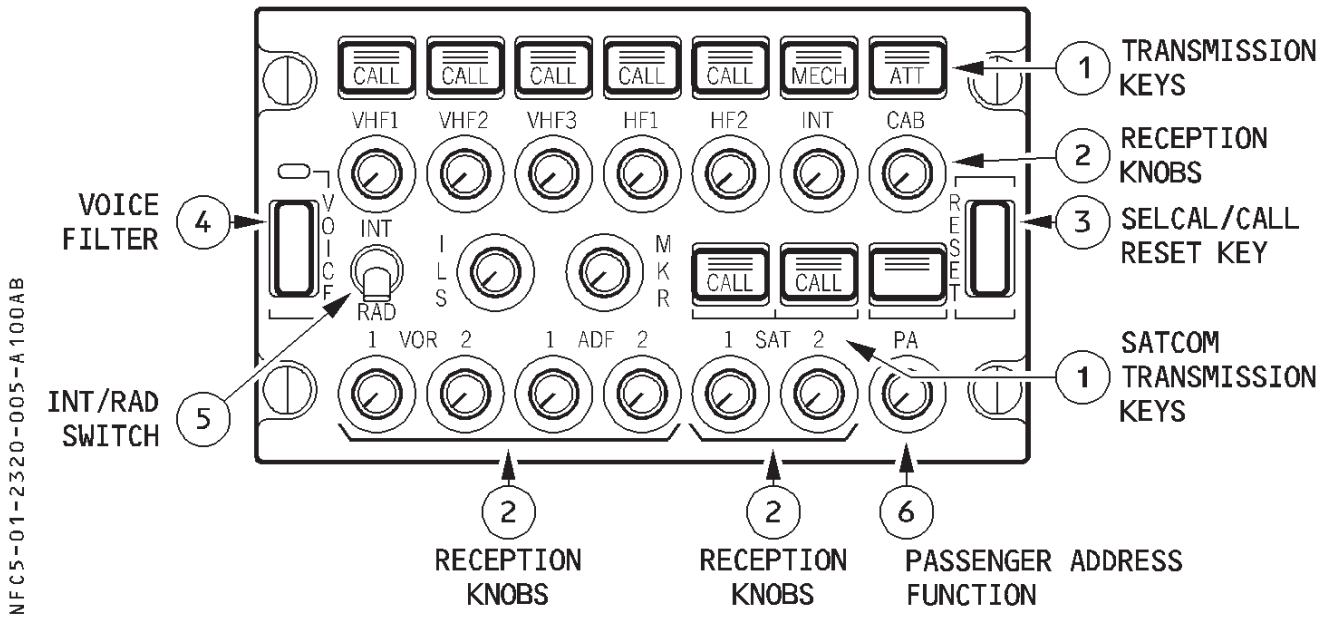
INT	:	Boom and mask mikes transmit on interphone regardless of which transmission key is selected. For reception on interphone, the crew member must have INT selected (INT reception knob out).
Neutral	:	Reception is normal. Boom and mask mikes do not transmit.
RAD (press and hold)	:	Boom and mask mikes transmit on the radio selected on the audio control panel.

⑥ Passenger address (PA) function

(Refer to PASSENGER ADDRESS 1.23.20, below).

CONTROLS

AUDIO CONTROL PANEL



(1) Transmission keys

Pressed	:	The associated channel is selected for transmission. The three green lines come on. The pilot deselects the channel by pressing the pushbutton again, or by selecting another channel.
CALL lt	:	The legend flashes amber (and buzzer sounds) when the SELCAL system detects a call.
MECH lt	:	The legend flashes amber (and buzzer sounds) for a call from the nose gear bay. MECH light goes off after 60 seconds, if it is not reset.
ATT lt	:	The legend flashes amber (and buzzer sounds) for a call from a cabin attendant. The ATT light goes off after 60 seconds, if it is not reset.
SAT CALL lt	:	This function is inoperative.

② Reception knobs

- Pressing and releasing the knob (knob out) selects the associated audio reception channel and the integral white light comes on.
- Rotating the knob adjusts the volume.
- The INTEG LT knob controls the brightness.
- Pressing the knob (knob stays in) disconnects the associated audio reception channel.

Note :
- *SATCOM reception knobs are inoperative.*
- *To receive DME audio navigation signals that are associated with an ILS or MLS station : The flight crew must not only select the ILS (or LS) pushbutton on the FCU, but also the VOR reception channel on the ACP. It is possible that the audio signals of the VOR, the ILS, and the DME (associated with ILS) could all sound at the same time.*

③ SELCAL/CALL RESET key

Pressing this key extinguishes CALL, MECH, and ATT lights.

④ ON VOICE key

This key allows the flight crew to inhibit the audio navigation signals (VOR, ADF)
Pressing this key filters out ident signals and turns on the green ON light.

⑤ INT/RAD sw

This switch operates as a press-to-talk switch for boom mike or oxygen mask mike.

INT	:	Boom and mask mikes transmit on interphone regardless of which transmission key is selected. For reception on interphone, the crew member must have INT selected (INT reception knob out).
Neutral	:	Reception is normal. Boom and mask mikes do not transmit.
RAD (press and hold)	:	Boom and mask mikes transmit on the radio selected on the audio control panel.

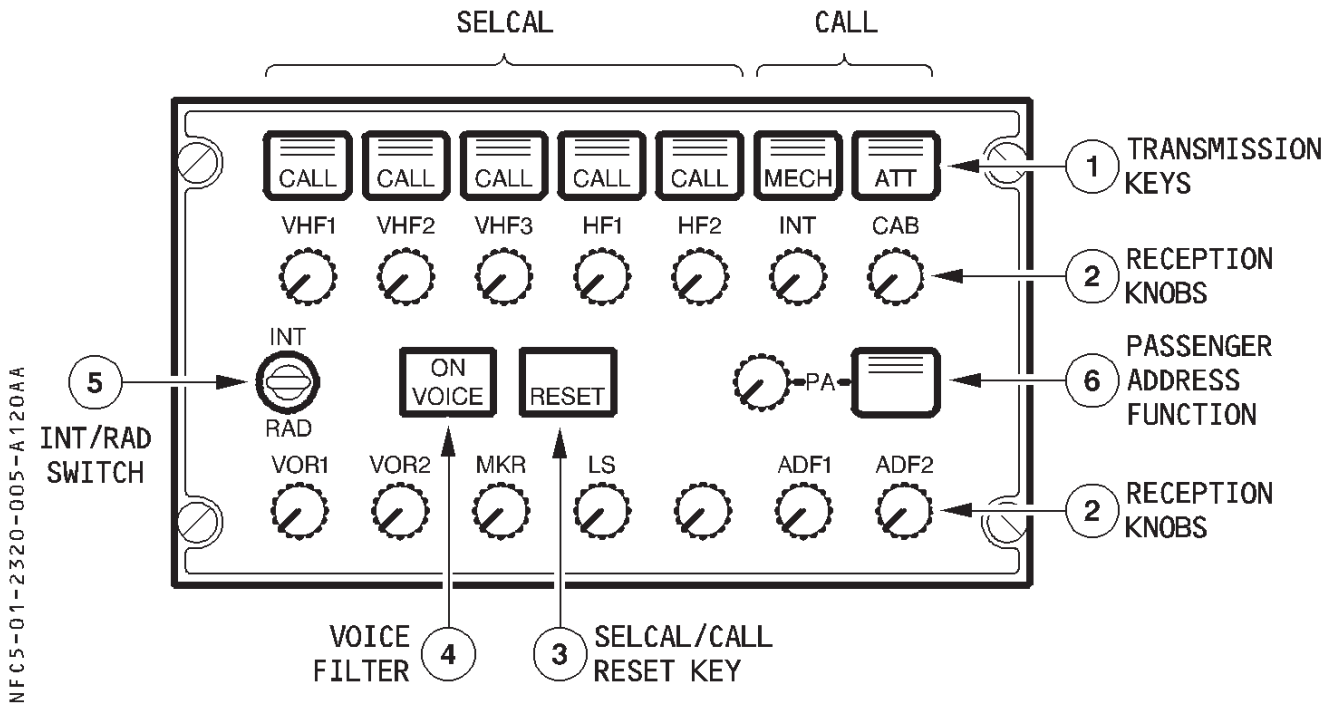
⑥ Passenger address (PA) function

(Refer to PASSENGER ADDRESS 1.23.20, below).

CONTROLS

AUDIO CONTROL PANEL

R



① Transmission keys

Pressed	:	The associated channel is selected for transmission. The three green lines come on. The pilot deselects the channel by pressing the pushbutton again, or selecting another channel.
CALL	:	The legend flashes in amber (and the buzzer sounds), when the SELCAL system detects a call.
MECH	:	The legend flashes in amber (and the buzzer sounds), for a call from the nose gear bay. The MECH light goes off after 60 seconds, if it is not reset.
ATT	:	The legend flashes in amber (and the buzzer sounds), for a call from a cabin crewmember. The ATT light goes off after 60 seconds, if it is not reset.

② Reception knobs

- Pressing and releasing the knob (knob out) selects the associated audio reception channel and the integral white light comes on.
- Rotating the knob adjusts the volume.
- The INTEG LT knob controls the brightness.
- Pressing the knob (knob stays in) disconnects the associated audio reception channel.

Note : For reception of DME audio navigation signals associated to an ILS or MLS station, the ILS (or LS) pushbutton on the FCU must also be selected.

③ SELCAL/CALL RESET key

Pressing this key extinguishes CALL, MECH, and ATT lights.

④ ON VOICE key

This key allows the flight crew to inhibit the audio navigation signals (VOR, ADF)
Pressing this key filters out ident signals and turns on the green ON light.

⑤ INT/RAD sw

This switch operates as a press-to-talk switch for boom mike or oxygen mask mike.

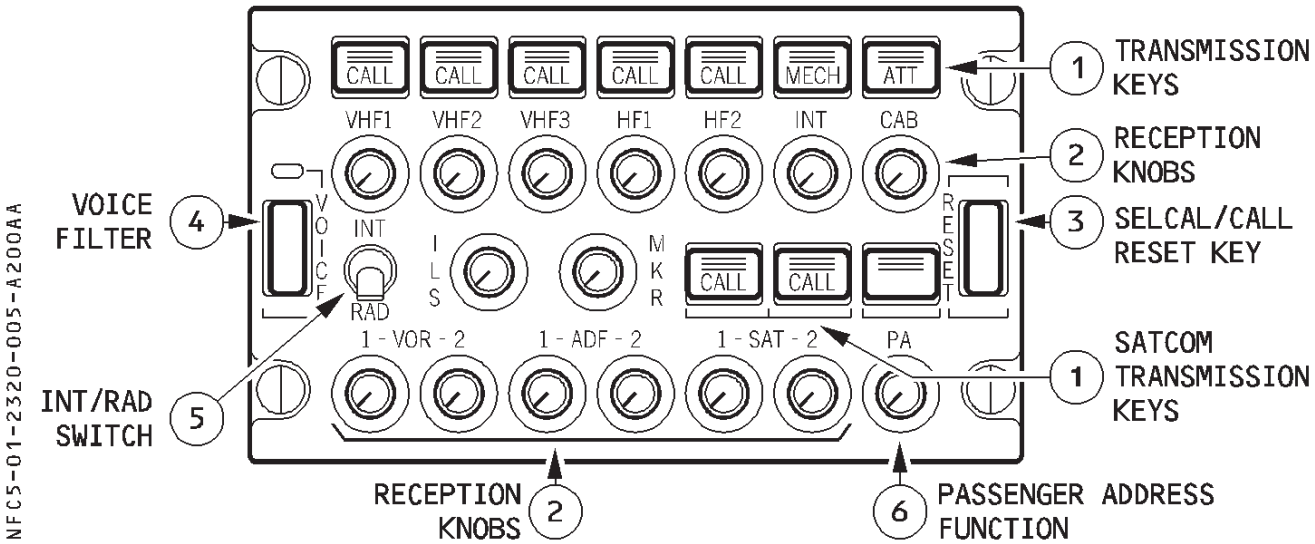
INT	:	Boom and mask mikes transmit on interphone regardless of which transmission key is selected. For reception on interphone, the crew member must have INT selected (INT reception knob out).
Neutral	:	Reception is normal. Boom and mask mikes do not transmit.
RAD (press and hold)	:	Boom and mask mikes transmit on the radio selected on the audio control panel.

⑥ Passenger address (PA) function

(Refer to PASSENGER ADDRESS 1.23.20, below).

CONTROLS

AUDIO CONTROL PANEL



(1) Transmission keys

Pressed	:	The associated channel is selected for transmission. The three green lines come on. The pilot deselects the channel by pressing the pushbutton again, or by selecting another channel.
CALL lt	:	The legend flashes amber (and buzzer sounds) when the SELCAL system detects a call.
MECH lt	:	The legend flashes amber (and buzzer sounds) for a call from the nose gear bay. The MECH light goes off after 60 seconds, if it is not reset.
ATT lt	:	The legend flashes amber (and buzzer sounds) for a call from a cabin attendant. The ATT light goes off after 60 seconds, if it is not reset.
SAT CALL lt	:	The legend flashes amber when the SATCOM system detects a call. The three green lines flash during the establishment of air to ground calls, or when SATCOM calls are on hold. After call establishment, the three green lines remain steady.

② Reception knobs

- Pressing and releasing the knob (knob out) selects the associated audio reception channel and the integral white light comes on.
- Rotating the knob adjusts the volume.
- R The INTEG LT knob controls the brightness.
- Pressing the knob (knob stays in) disconnects the associated audio reception channel.

R *Note : To receive DME audio navigation signals that are associated with an ILS or MLS*
R *station : The flight crew must not only select the ILS (or LS) pushbutton on the*
R *FCU, but also the VOR reception channel on the ACP. It is possible that the audio*
R *signals of the VOR, the ILS, and the DME (associated with ILS) could all sound*
R *at the same time.*

③ SELCAL/CALL RESET key

Pressing this key extinguishes CALL, MECH, and ATT lights.

④ ON VOICE key

This key allows the flight crew to inhibit the audio navigation signals (VOR, ADF)
Pressing this key filters out ident signals and turns on the green ON light.

⑤ INT/RAD sw

This switch operates as a press-to-talk switch for boom mike or oxygen mask mike.

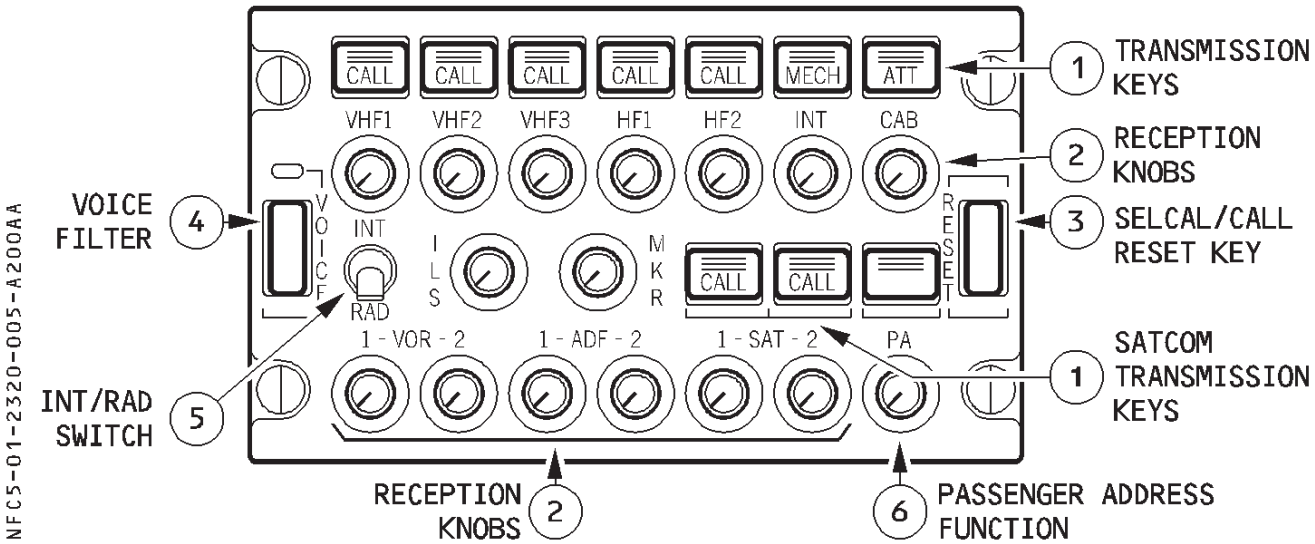
INT	:	Boom and mask mikes transmit on interphone regardless of which transmission key is selected. For reception on interphone, the crew member must have INT selected (INT reception knob out).
Neutral	:	Reception is normal. Boom and mask mikes do not transmit.
RAD (press and hold)	:	Boom and mask mikes transmit on the radio selected on the audio control panel.

⑥ Passenger address (PA) function

(Refer to PASSENGER ADDRESS 1.23.20, below).

CONTROLS

AUDIO CONTROL PANEL



(1) Transmission keys

Pressed	:	The associated channel is selected for transmission. The three green lines come on. The pilot deselects the channel by pressing the pushbutton again, or by selecting another channel.
CALL lt	:	The legend flashes amber (and buzzer sounds) when the SELCAL system detects a call.
MECH lt	:	The legend flashes amber (and buzzer sounds) for a call from the nose gear bay. The MECH light goes off after 60 seconds, if it is not reset.
ATT lt	:	The legend flashes amber (and buzzer sounds) for a call from a cabin attendant. The ATT light goes off after 60 seconds, if it is not reset.
SAT CALL lt	:	The legend flashes amber when the SATCOM system detects a call. The three green lines flash during the establishment of air to ground calls, or when SATCOM calls are on hold. After call establishment, the three green lines remain steady.

② Reception knobs

- Pressing and releasing the knob (knob out) selects the associated audio reception channel and the integral white light comes on.
- Rotating the knob adjusts the volume.
- The INTEG LT knob controls the brightness.
- Pressing the knob (knob stays in) disconnects the associated audio reception channel.

Note : For reception of DME audio navigation signals associated to an ILS or MLS station, the ILS (or LS) pushbutton on the FCU must also be selected.

③ SELCAL/CALL RESET key

Pressing this key extinguishes CALL, MECH, and ATT lights.

④ ON VOICE key

This key allows the flight crew to inhibit the audio navigation signals (VOR, ADF)
Pressing this key filters out ident signals and turns on the green ON light.

⑤ INT/RAD sw

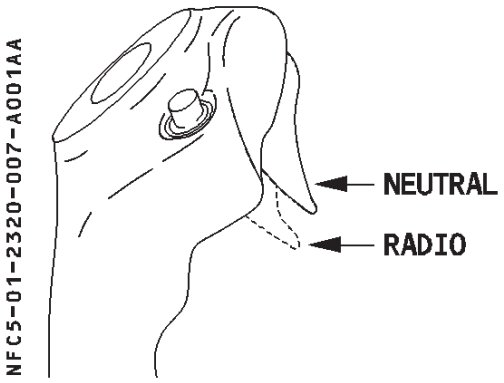
This switch operates as a press-to-talk switch for boom mike or oxygen mask mike.

INT	:	Boom and mask mikes transmit on interphone regardless of which transmission key is selected. For reception on interphone, the crew member must have INT selected (INT reception knob out).
Neutral	:	Reception is normal. Boom and mask mikes do not transmit.
RAD (press and hold)	:	Boom and mask mikes transmit on the radio selected on the audio control panel.

⑥ Passenger address (PA) function

(Refer to PASSENGER ADDRESS 1.23.20, below).

SIDE STICK RADIO SELECTOR



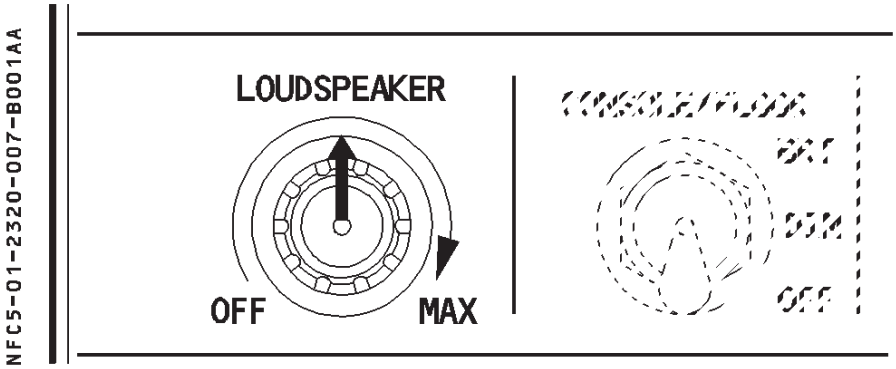
This selector has the same function as the INT/RAD switch on the ACP.

NEUTRAL (spring-loaded) : Boom and mask mikes are dead.
Reception is normal.

RADIO (squeezed) : Boom and mask mikes transmit through the equipment selected by the transmission key on the ACP.

Note : If RADIO is selected on the side stick when the INT/RAD switch is on INT, the radio function has priority over the interphone function.

LOUDSPEAKER VOLUME KNOB



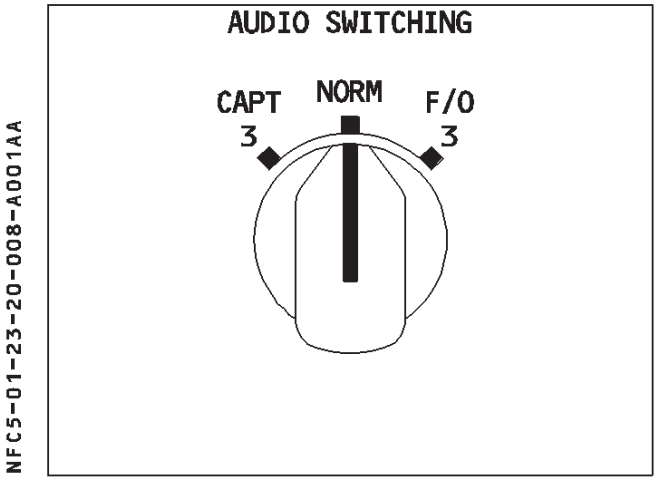
This knob adjusts the volume of the loudspeaker for radio communication.

OFF : Loudspeaker does not respond to signals from the aircraft's radio equipment.

Clockwise rotation : Loudspeaker broadcasts signals from the aircraft's radio equipment at increasing volume.

Note : This knob does not control the loudness of aural alert and voice messages.

AUDIO SWITCHING



The crew can switch to the third ACP if ACP1 or ACP2 fails.
When the crew does this, it takes away the third occupant's access to the acoustic equipment. AUDIO 3 XFRD appears in green on the ECAM MEMO display.

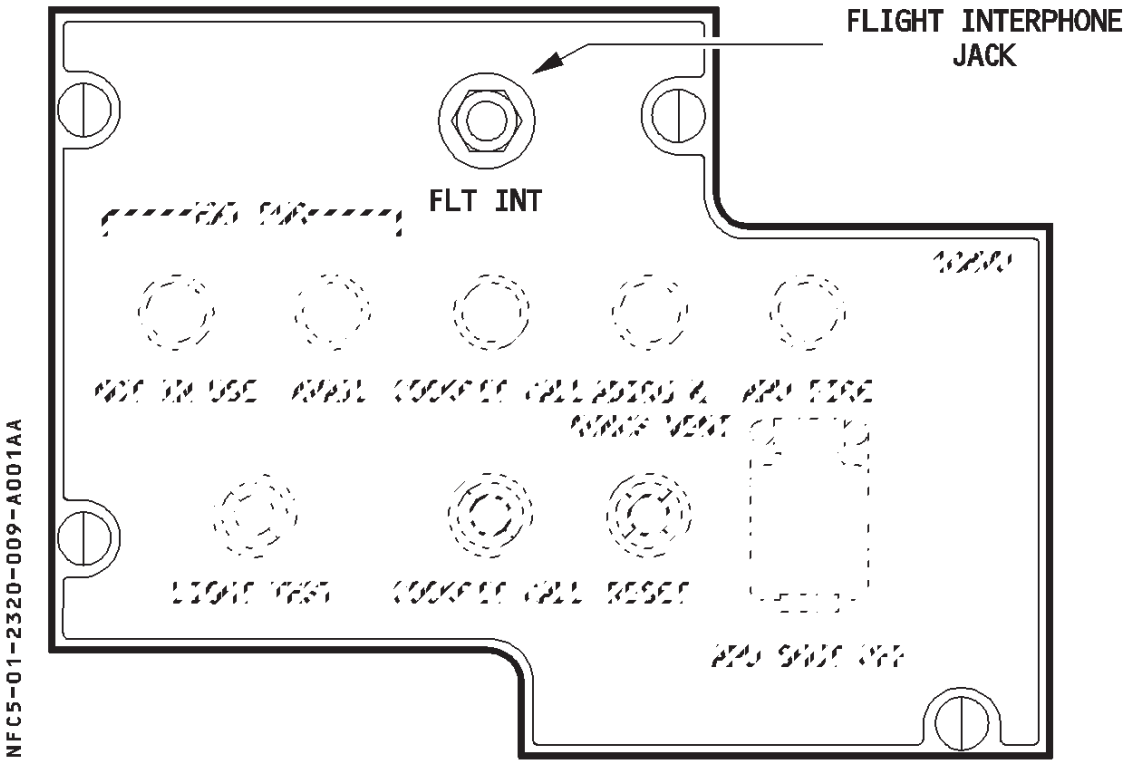
NORM : Each crew member uses his dedicated communication equipment.
CAPT 3 : The pilot uses his acoustic equipment and the third occupant's ACP.
F/O 3 : The copilot uses his acoustic equipment and the third occupant's ACP.

INTERPHONE SYSTEMS

FLIGHT INTERPHONE SYSTEM

This system allows the flight crew members to communicate among themselves and, through a jack on the external power panel, with the ground mechanic.

EXTERNAL POWER PANEL (FORWARD OF THE NOSE L/G BAY)



COCKPIT OPERATION FOR GROUND MECHANIC COMMUNICATION

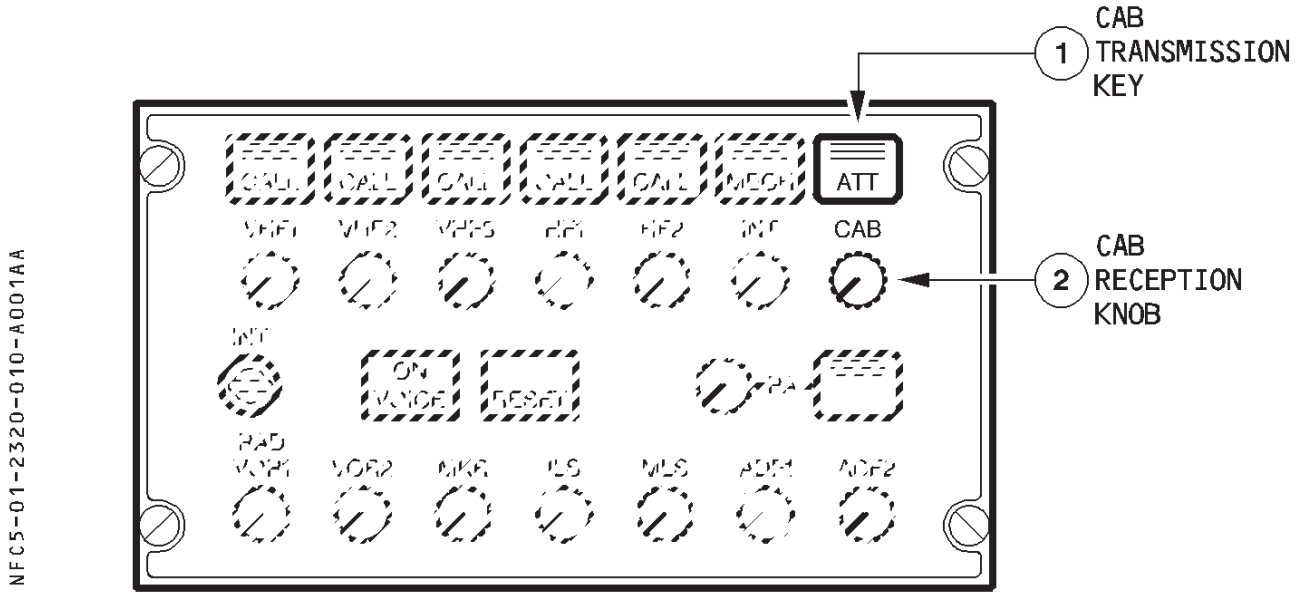
R

	MECH TRANSMISSION KEY ON ACP	INT RECEPTION KNOB ON ACP	INT/RAD SW ON ACP	PUSH TO TALK ON HANDMIKE
BOOMSET OR OXYGEN MASK	PRESSED	OUT	INT OR RAD	–
HANDMIKE	PRESSED	OUT	–	PRESSED

CABIN INTERPHONE SYSTEM

The system allows the flight crew to communicate with the flight attendants, and the flight attendants to communicate among themselves.

R



(1) CAB transmission key

Pressed : Three green lines come on.
Boom, mask, and hand mikes may be used for cabin interphone.

(2) CAB reception knob

Pressed and released (knob out) : The integral white light comes on.
The station receives audio signals from the cabin.
Rotating the knob adjusts the volume.

Pressed (knob in) : The white light goes out.
The cabin interphone is disconnected.

COCKPIT OPERATION FOR CABIN COMMUNICATION

	CAB TRANSMISSION KEY ON ACP	CAB RECEPTION KNOB ON ACP	INT/RAD SW ON ACP	PUSH TO TALK ON HANDMIKE
BOOMSET OR OXYGEN MASK	PRESSED	OUT	RAD	–
HANDMIKE	PRESSED	OUT	–	PRESSED

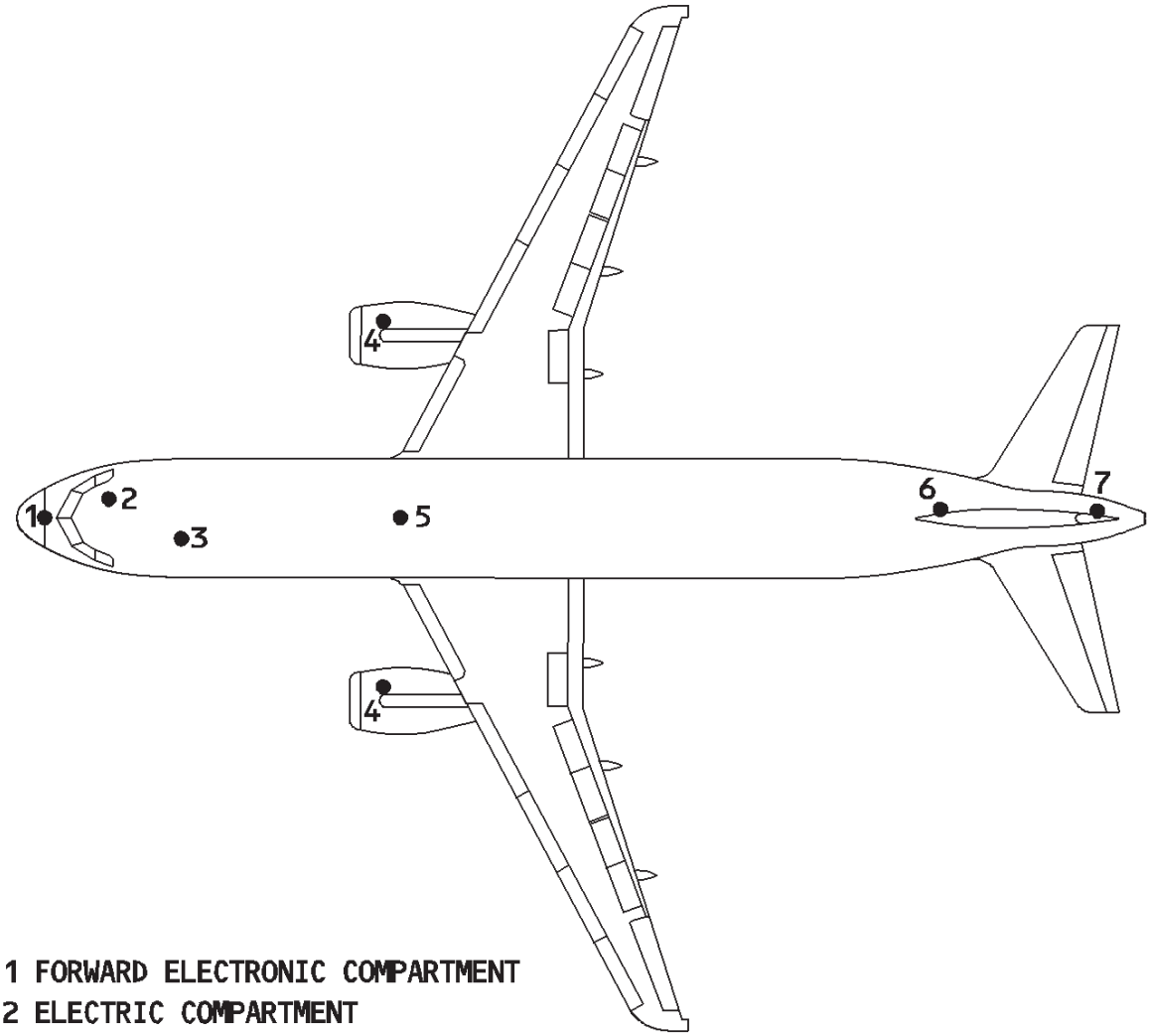
SERVICE INTERPHONE SYSTEM

The system allows for communication between :
- the flight crew and the service interphone jacks
- the flight attendant stations and the service interphone jacks
- the different service interphone jacks.

The Service Interphone system has :
- seven interphone jacks
- an OVRD switch located on the overhead panel.

The audio lines from the interphone jacks are connected to both CIDS directors.

LOCATION OF INTERPHONE JACKS



1 FORWARD ELECTRONIC COMPARTMENT
2 ELECTRIC COMPARTMENT
3 AVIONICS COMPARTMENT
4 ENGINE (L & R)
5 BELLY FAIRING PANEL
6 AFT FUSELAGE
7 NEAR APU BAY

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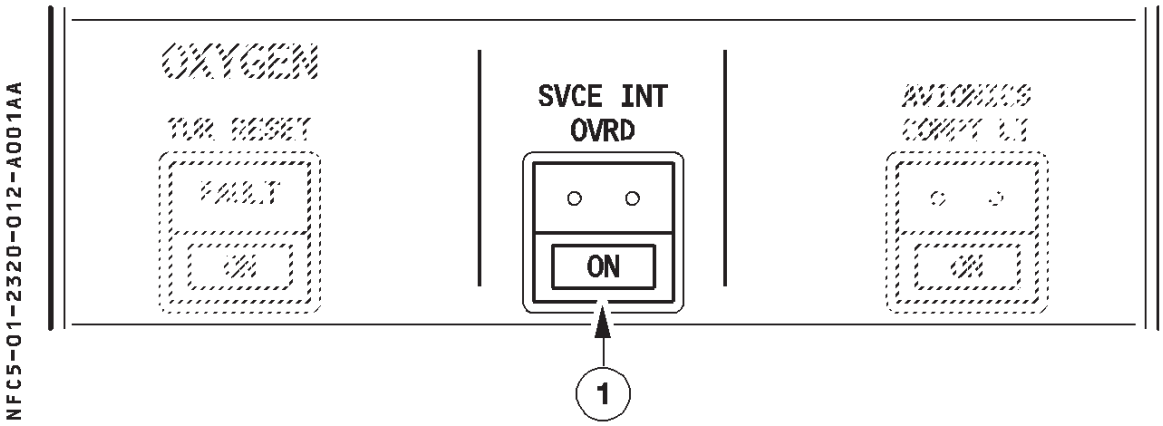
CONTROLS AND INDICATORS AT OVERHEAD PANEL



(1) SVCE INT OVRD pb sw

Auto : Ground personnel can communicate with the flight crew by means of the service interphone jacks 10 seconds after the aircraft has landed. The landing gear must be compressed.

ON : Communication is possible when the landing gear is not compressed. The ON light is white.

COCKPIT OPERATION

R

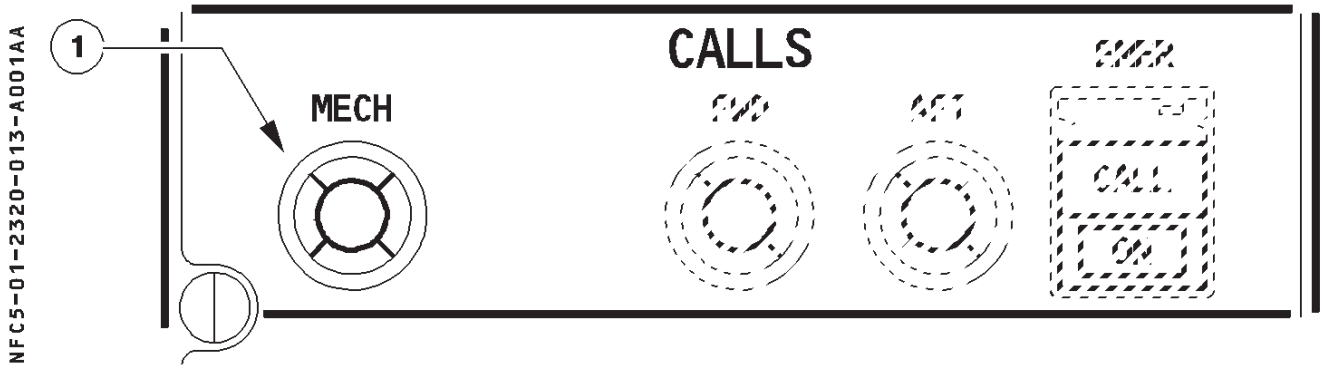
	CAB TRANSMISSION KEY ON ACP	CAB RECEPTION KNOB ON ACP	INT/RAD SW ON ACP	PUSH TO TALK ON HANDMIKE	SVCE INT OVRD PB SW
BOOMSET	PRESSED	OUT	RAD	–	SEE ABOVE CONDITIONS.
HANDMIKE	PRESSED	OUT	–	PRESSED	

CALL SYSTEMS

GROUND MECHANIC CALL

The system allows the flight crew and ground mechanics to communicate with each other.

CONTROLS AND INDICATORS ON OVERHEAD PANEL

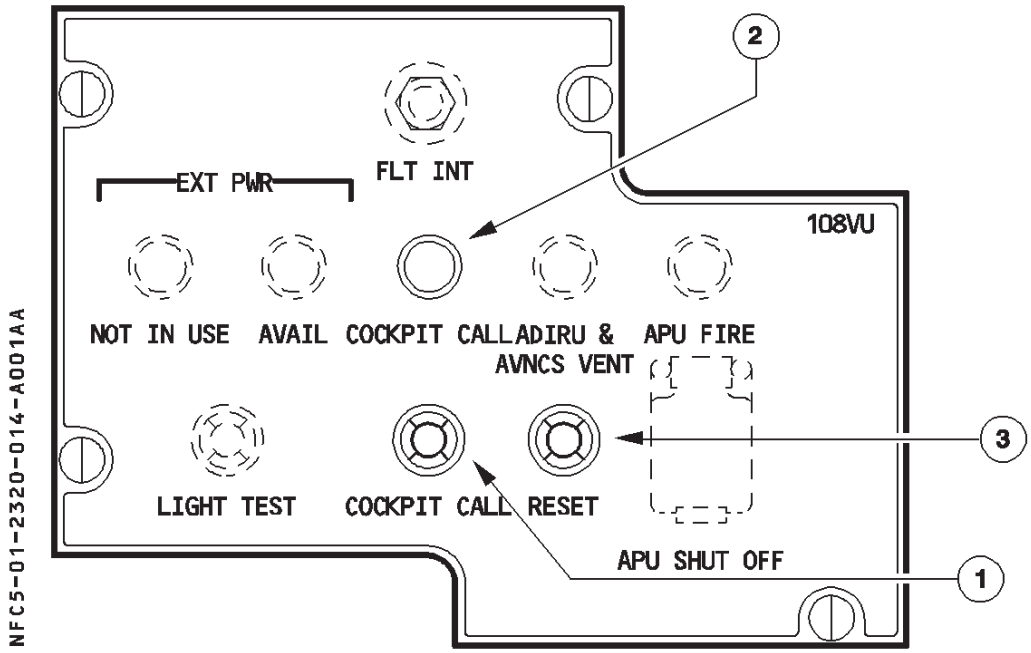


(1) MECH pb

Pressed (and held) : COCKPIT CALL lights up blue on the external power panel in the nose L/G bay.
An external horn sounds.

Released : COCKPIT CALL remains lighted.
The ground mechanic can extinguish it by pressing the RESET button on the external power panel. The external horn stops sounding.

R CONTROLS AND INDICATORS ON THE EXTERNAL POWER PANEL



(1) COCKPIT CALL pb

Pressed : This calls the cockpit.
The MECH lights flash amber on the ACPs and a buzzer sounds.

Released : The MECH lights go out after 60 seconds if they are not reset on the ACPs.
The buzzer stops.

(2) COCKPIT CALL lt

The blue light appears when cockpit calls the ground mechanic. An external horn also sounds.

(3) RESET pb

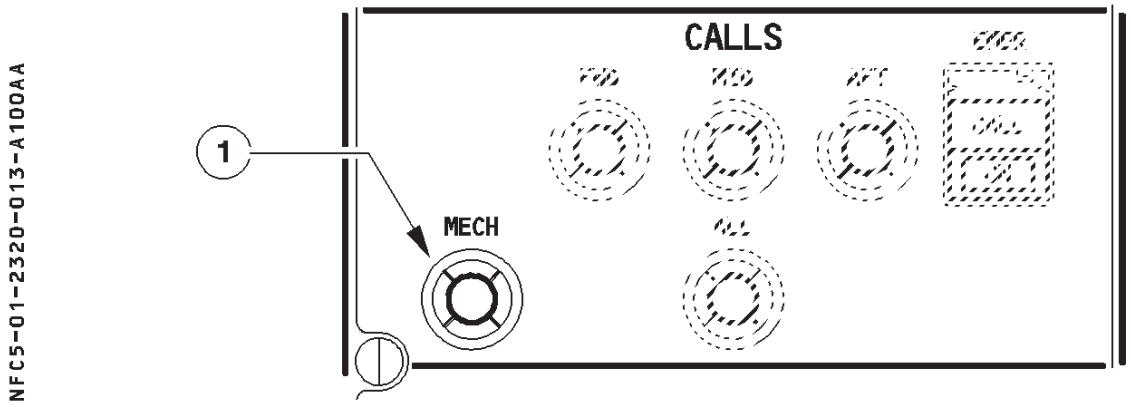
R Pressed : The COCKPIT CALL light goes out.

CALL SYSTEMS

GROUND MECHANIC CALL

The system allows the flight crew and ground mechanics to communicate with each other.

CONTROLS AND INDICATORS ON OVERHEAD PANEL

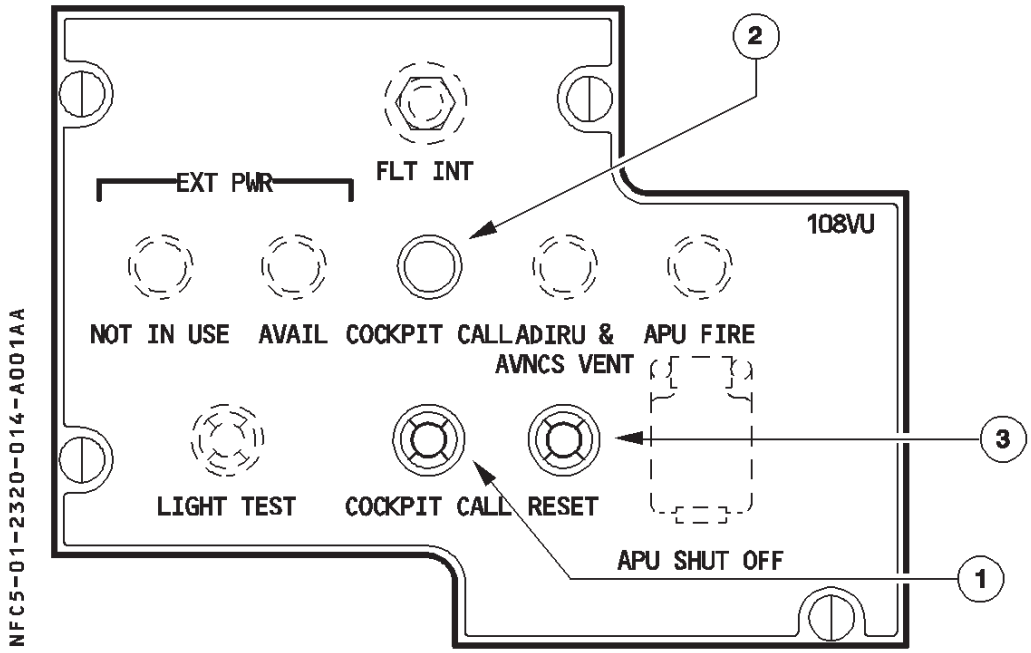


(1) MECH pb

Pressed (and held) : COCKPIT CALL lights up blue on the external power panel in the nose L/G bay.
An external horn sounds.

Released : COCKPIT CALL remains lighted.
The ground mechanic can extinguish it by pressing the RESET button on the external power panel. The external horn stops sounding.

R CONTROLS AND INDICATORS ON THE EXTERNAL POWER PANEL



(1) COCKPIT CALL pb

Pressed : This calls the cockpit.
The MECH lights flash amber on the ACPs and a buzzer sounds.

Released : The MECH lights go out after 60 seconds if they are not reset on the ACPs.
The buzzer stops.

(2) COCKPIT CALL lt

The blue light appears when cockpit calls the ground mechanic. An external horn also sounds.

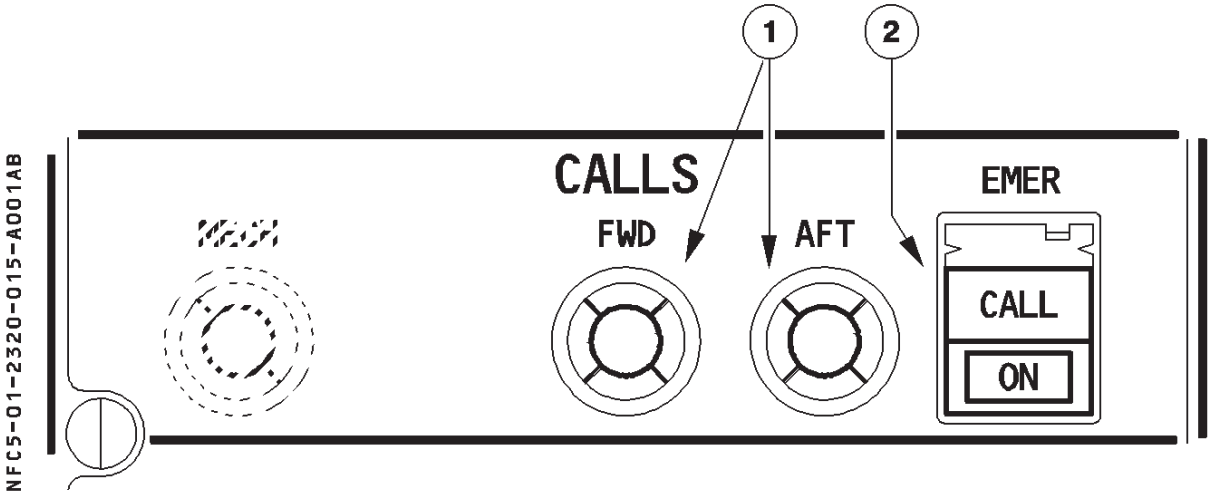
(3) RESET pb

R Pressed : The COCKPIT CALL light goes out.

CABIN CALL SYSTEM

The cabin call system is for communication between the cockpit and the cabin.

CALLS FROM THE COCKPIT



(1) FWD/AFT pushbutton

When pressed, the following aural and visual alerts will trigger in the cabin :

R – Two lights come on in pink on the FWD or AFT area call panel, as applicable.
– On the Attendant Indication Panel (AIP), the "CAPTAIN CALL" message appears and a light comes on in green.
– A high-low chime sounds in the FWD or AFT section of the cabin, as applicable.

(2) EMER pushbutton-switch (guarded)

When pressed, the following aural and visual alerts will trigger in the cabin :
– Two pink lights flash on all area call panels.
– The "EMERGENCY CALL" message appears on all AIPs.
– A high-low chime sounds three times, on all of the loudspeakers.

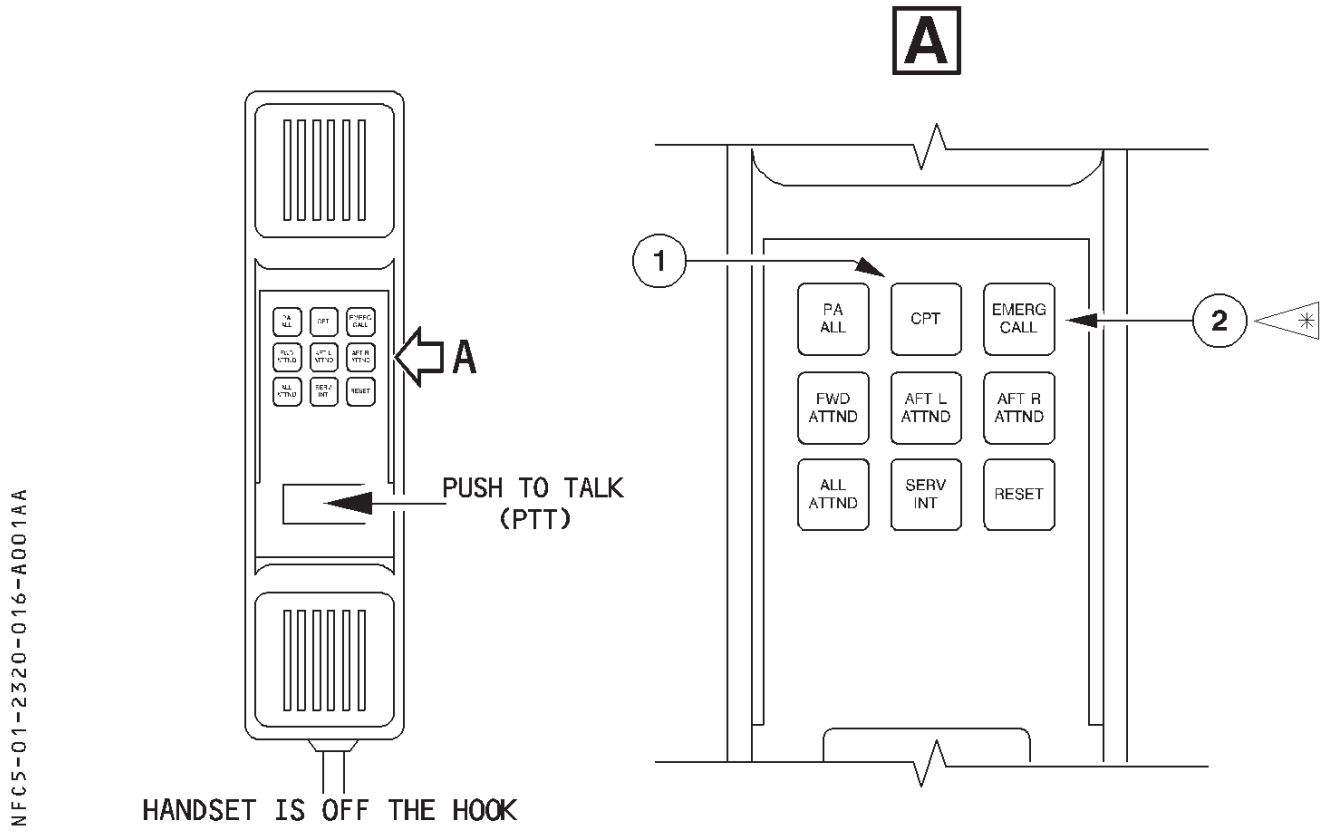
Also, on the cockpit CALLS panel, the ON light flashes in white and the CALL light flashes in amber.

R If the cabin handset has an EMER CALL key, the following aural and visual alerts will
R trigger in the cockpit, when an emergency call is made from the cabin to the cockpit :
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all Audio Control Panels (ACPs).
– Three buzzers will sound consecutively (for approx. three seconds each).

The cabin call system will reset, when the cabin crewmember hangs up the handset.

CALLS FROM THE CABIN

R



(1) CPT key

When pressed, the following aural and visual alerts will trigger in the cockpit :
– The ATT lights will flash on all Audio Control Panels (ACPs).
– A buzzer (inhibited during takeoff and landing) will sound.
In the cabin, the "CAPTAIN" message will appear on the Attendant Indication Panel (AIP) for which the CPT key was pressed.

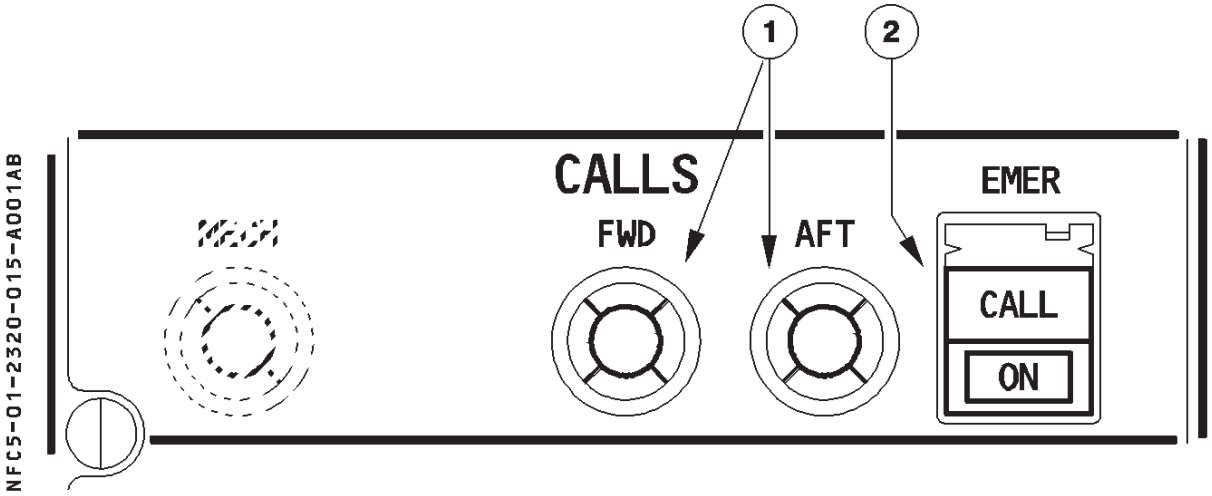
R (2) EMER CALL key ⊲

When pressed, the following aural and visual alerts will trigger in the cockpit :
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all ACPs
R – Three buzzers (inhibited during takeoff and landing) will sound consecutively.
In the cabin, the "EMERGENCY CALL" message will appear on all AIPs.

CABIN CALL SYSTEM

The cabin call system is for communication between the cockpit and the cabin.

CALLS FROM THE COCKPIT



(1) FWD/AFT pushbutton

When pressed, the following aural and visual alerts will trigger in the cabin :

R – Two lights come on in pink on the FWD or AFT area call panel, as applicable.
– On the Attendant Indication Panel (AIP), the "CAPTAIN CALL" message appears and a light comes on in green.
– A high-low chime sounds in the FWD or AFT section of the cabin, as applicable.

(2) EMER pushbutton-switch (guarded)

When pressed, the following aural and visual alerts will trigger in the cabin :
– Two pink lights flash on all area call panels.
– The "EMERGENCY CALL" message appears on all AIPs.
– A high-low chime sounds three times, on all of the loudspeakers.

Also, on the cockpit CALLS panel, the ON light flashes in white and the CALL light flashes in amber.

R If the cabin handset has an EMER CALL key, the following aural and visual alerts will
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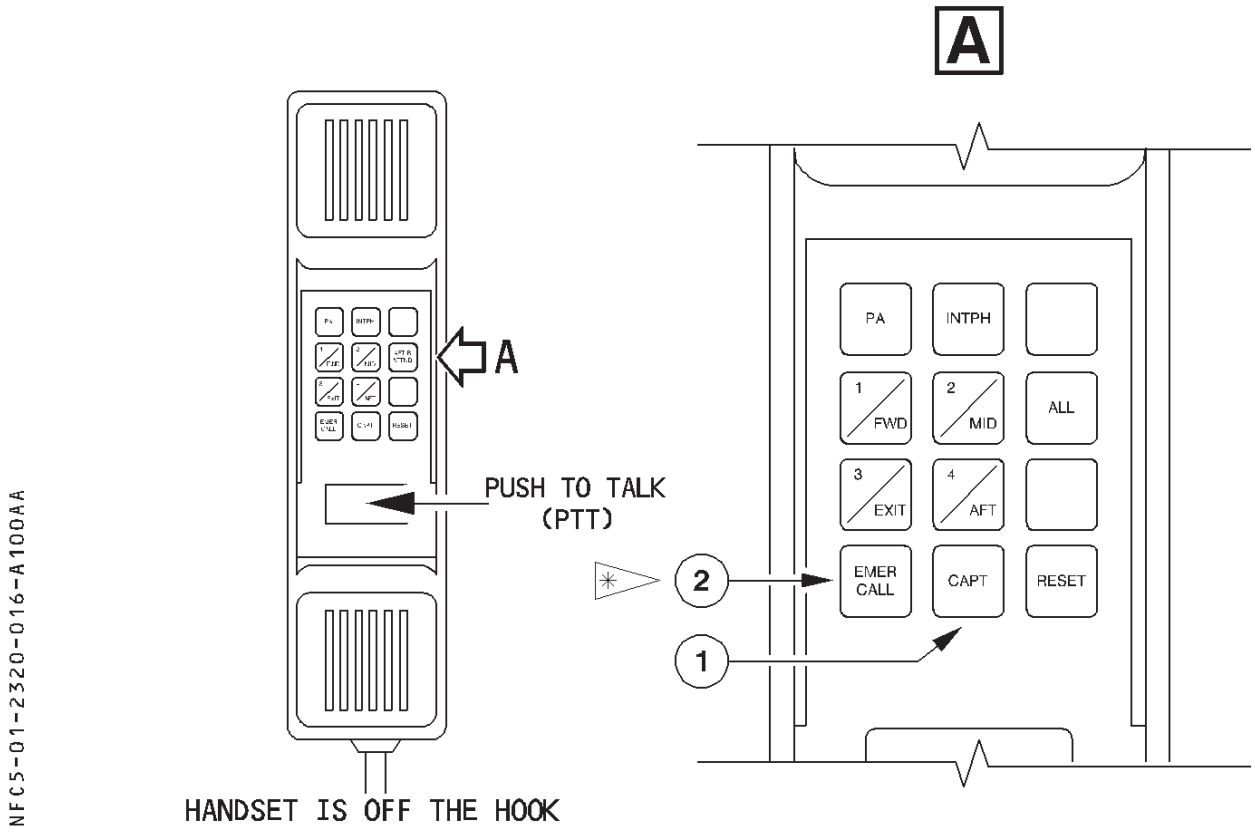
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all Audio Control Panels (ACPs).
– Three buzzers will sound consecutively (for approx. three seconds each).

The cabin call system will reset, when the cabin crewmember hangs up the handset.

CALLS FROM THE CABIN

R



① CAPT key

When pressed, the following aural and visual alerts will trigger in the cockpit :
– The ATT lights will flash on all Audio Control Panels (ACPs).
– A buzzer (inhibited during takeoff and landing) will sound.
In the cabin, the "CAPTAIN" message will appear on the Attendant Indication Panel (AIP) for which the CAPT key was pressed.

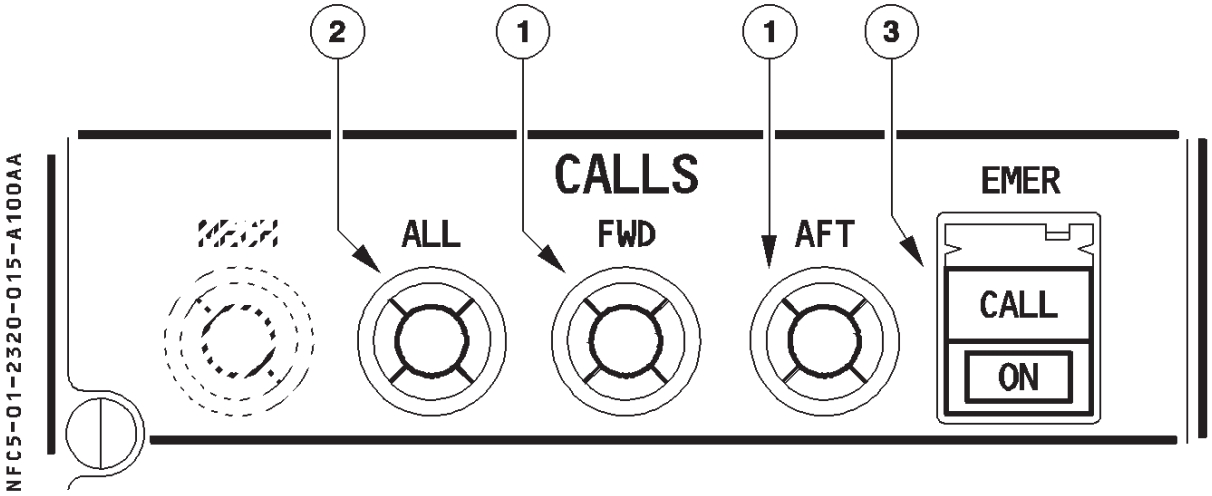
R ② EMER CALL key ⊲

When pressed, the following aural and visual alerts will trigger in the cockpit :
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all ACPs
R – Three buzzers (inhibited during takeoff and landing) will sound consecutively.
In the cabin, the "EMERGENCY CALL" message will appear on all AIPs.

CABIN CALL SYSTEM

The cabin call system is for communication between the cockpit and the cabin.

CALLS FROM THE COCKPIT



(1) FWD/AFT pushbutton

When pressed, the following aural and visual alerts will trigger in the cabin :
R – Two lights come on in pink on the FWD or AFT area call panel, as applicable.
– On the Attendant Indication Panel (AIP), the "CAPTAIN CALL" message appears and a light comes on in green.
– A high-low chime sounds in the FWD or AFT section of the cabin, as applicable.

(2) ALL pushbutton

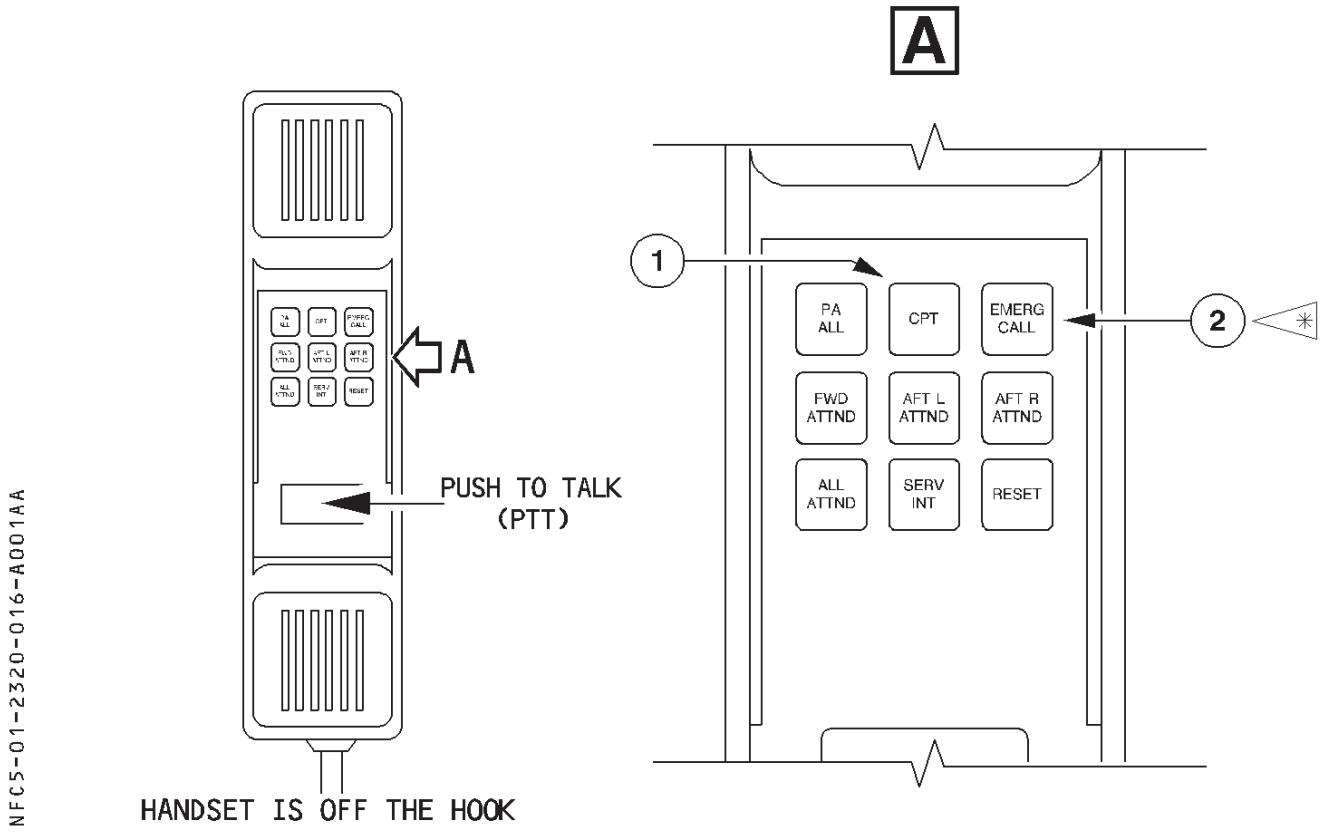
When pressed, all the stations simultaneously respond, as indicated above.

(3) EMER pushbutton-switch (guarded)

When pressed, the following aural and visual alerts will trigger in the cabin :
– Two pink lights flash on all area call panels.
– The "EMERGENCY CALL" message appears on all AIPs.
– A high-low chime sounds three times, on all of the loudspeakers.
Also, on the cockpit CALLS panel, the ON light flashes in white and the CALL light flashes in amber.
R If the cabin handset has an EMER CALL key, the following aural and visual alerts will
R trigger in the cockpit, when an emergency call is made from the cabin to the cockpit :
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all Audio Control Panels (ACPs).
– Three buzzers will sound consecutively (for approx. three seconds each).
The cabin call system will reset, when the cabin crewmember hangs up the handset.

CALLS FROM THE CABIN

R



(1) CPT key

When pressed, the following aural and visual alerts will trigger in the cockpit :
– The ATT lights will flash on all Audio Control Panels (ACPs).
– A buzzer (inhibited during takeoff and landing) will sound.
In the cabin, the "CAPTAIN" message will appear on the Attendant Indication Panel (AIP) for which the CPT key was pressed.

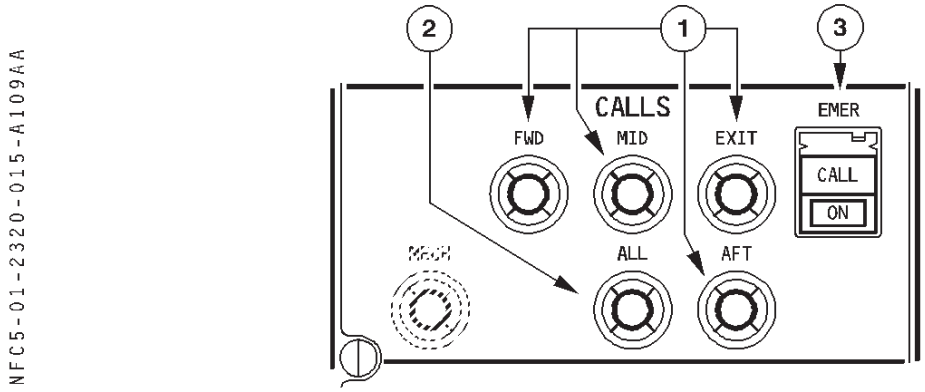
R (2) EMER CALL key ⊲

When pressed, the following aural and visual alerts will trigger in the cockpit :
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all ACPs
R – Three buzzers (inhibited during takeoff and landing) will sound consecutively.
In the cabin, the "EMERGENCY CALL" message will appear on all AIPs.

CABIN CALL SYSTEM

The cabin call system is for communication between the cockpit and the cabin.

CALLS FROM THE COCKPIT



① FWD/MID/EXIT/AFT pushbuttons

When pressed, the following aural and visual alerts will trigger in the cabin :

R – Two lights come on in pink on the FWD/MID/EXIT/AFT area call panel, as applicable.
– On the Attendant Indication Panel (AIP), the "CAPTAIN CALL" message appears and a light comes on in green.
– A high-low chime sounds in the FWD/MID/EXIT/AFT section of the cabin, as applicable.

② ALL pushbutton

When pressed, all the stations simultaneously respond, as indicated above.

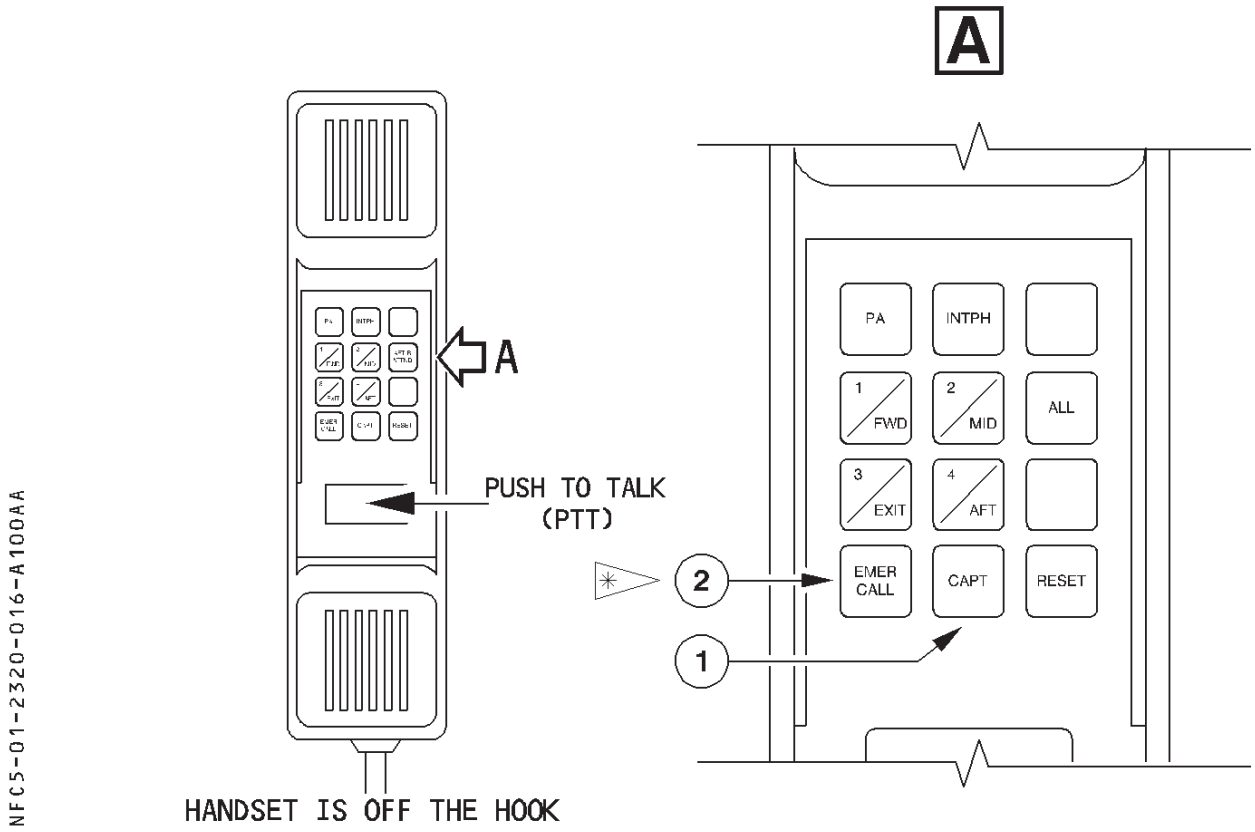
③ EMER pushbutton-switch (guarded)

When pressed, the following aural and visual alerts will trigger in the cabin :
– Two pink lights flash on all area call panels.
– The "EMERGENCY CALL" message appears on all AIPs.
– A high-low chime sounds three times, on all of the loudspeakers.
Also, on the cockpit CALLS panel, the ON light flashes in white and the CALL light flashes in amber.

R If the cabin handset has an EMER CALL key, the following aural and visual alerts will
R trigger in the cockpit, when an emergency call is made from the cabin to the cockpit :
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all Audio Control Panels (ACPs).
– Three buzzers will sound consecutively (for approx. three seconds each).
The cabin call system will reset, when the cabin crewmember hangs up the handset.

CALLS FROM THE CABIN

R



① CAPT key

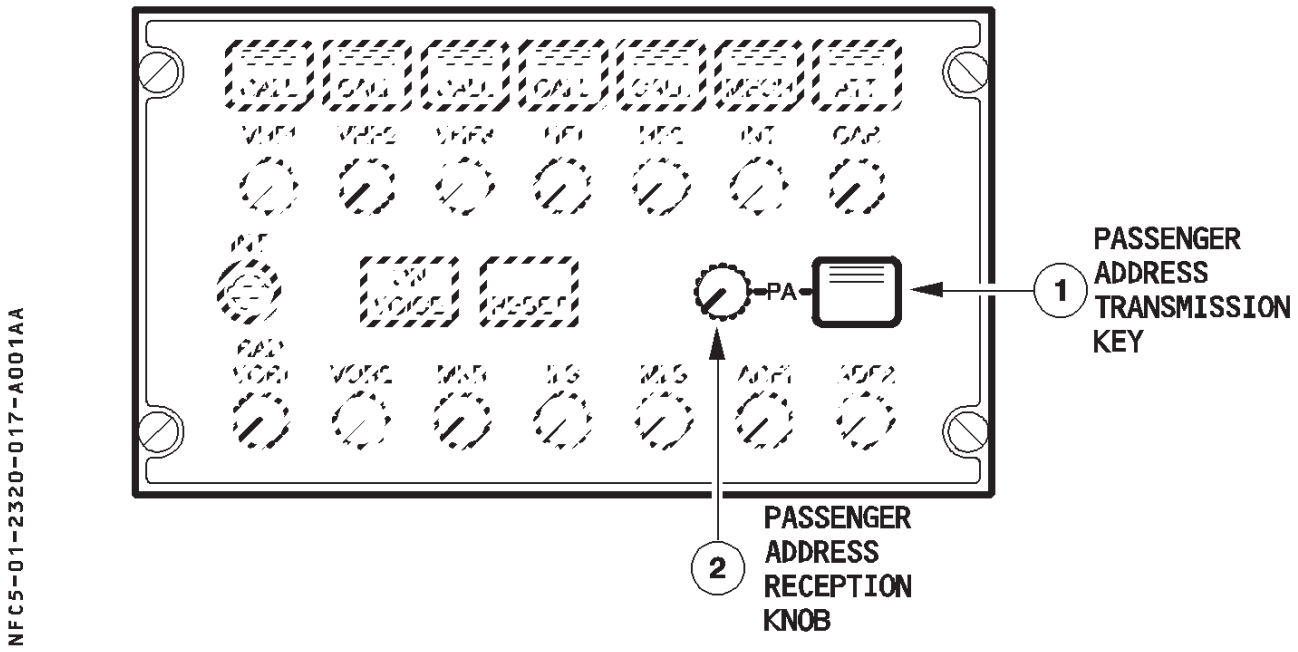
When pressed, the following aural and visual alerts will trigger in the cockpit :
– The ATT lights will flash on all Audio Control Panels (ACPs).
– A buzzer (inhibited during takeoff and landing) will sound.
In the cabin, the "CAPTAIN" message will appear on the Attendant Indication Panel (AIP) for which the CAPT key was pressed.

R ② EMER CALL key ⊲

When pressed, the following aural and visual alerts will trigger in the cockpit :
– On the EMER pb-sw : The ON light flashes in white, and the CALL light flashes in amber.
– The ATT lights will flash on all ACPs
R – Three buzzers (inhibited during takeoff and landing) will sound consecutively.
In the cabin, the "EMERGENCY CALL" message will appear on all AIPs.

PASSENGER ADDRESS

The passenger address allows flight personnel to make announcements to passengers in the cabin through loudspeakers. It can be operated from the cockpit (with ACP or handset) or from the cabin (attendant stations).



(1) PA transmission key

Pressed and held : The flight crew may use a boom, mask, or hand mike to make an announcement.
Three green lines come on.

Note : The flight crew may use a cockpit handset to make PA announcements without action on the ACPs.

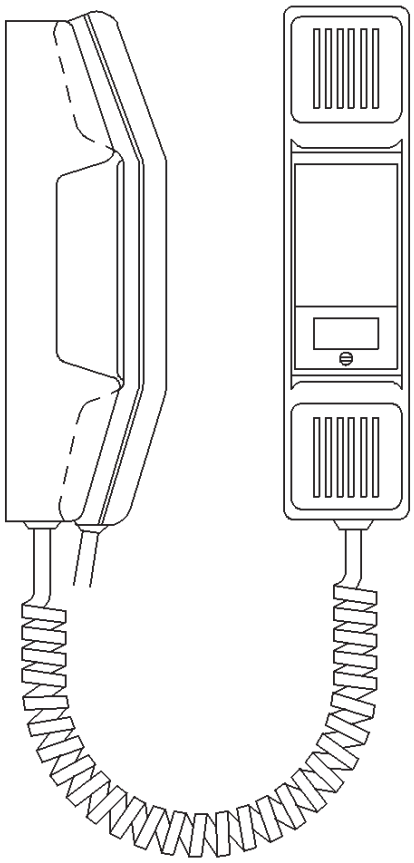
(2) PA reception knob

Pressed and released (knob out) : The message goes to the loudspeakers and the integral white light comes on.
R The flight crew can rotate the knob to adjust the volume.

Pressed (knob in) : The PA system is disconnected.
The white light goes out.

Cockpit handset

The cockpit handset at the bottom of the pedestal is for PA announcements.



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PA from cockpit

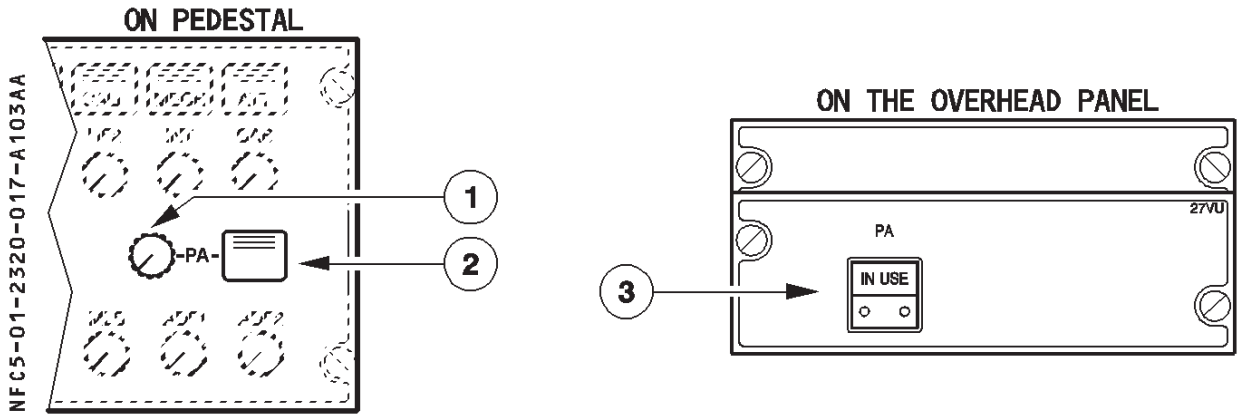
R

	PA TRANSMISSION KEY ON ACP	PA RECEPTION KNOB ON ACP	PUSH TO TALK ON HANDMIKE	PUSH TO TALK ON HANDSET
BOOMSET OR OXYGEN MASK	PRESSED	OUT		
HANDMIKE	PRESSED	OUT	PRESSED	
HANDSET				PRESSED

PASSENGER ADDRESS

The passenger address allows flight personnel to make announcements to passengers in the cabin through loudspeakers. It can be operated from the cockpit (with ACP or handset) or from the cabin (attendant stations).

CONTROLS and INDICATORS



① PA transmission key

Pressed and held : The flight crew may use a boom, mask, or hand mike to make an announcement.
Three green lines come on.

Note : The flight crew may use a cockpit handset to make PA announcements without action on the ACPs.

② PA reception knob

Pressed and released (knob out) : The message goes to the loudspeakers and the integral white light comes on.
R The flight crew can rotate the knob to adjust the volume.

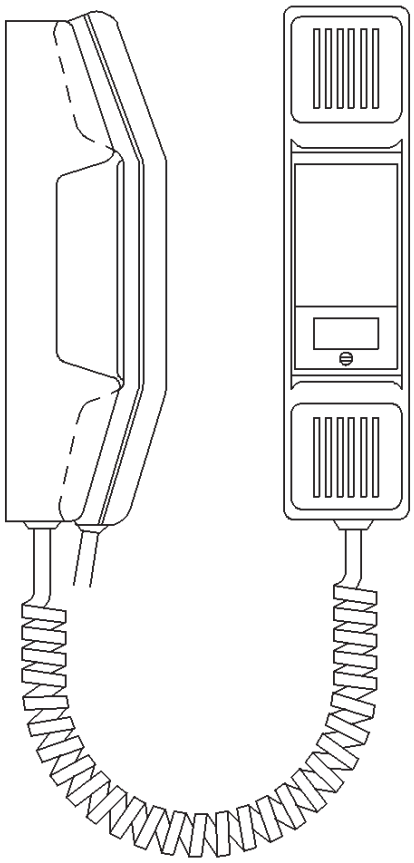
Pressed (knob in) : The PA system is disconnected.
The white light goes out.

③ PA IN USE lt

Comes on white when the PA is activated from the cockpit or from the cabin (cabin attendant or prerecorded announcement).

Cockpit handset

The cockpit handset at the bottom of the pedestal is for PA announcements.



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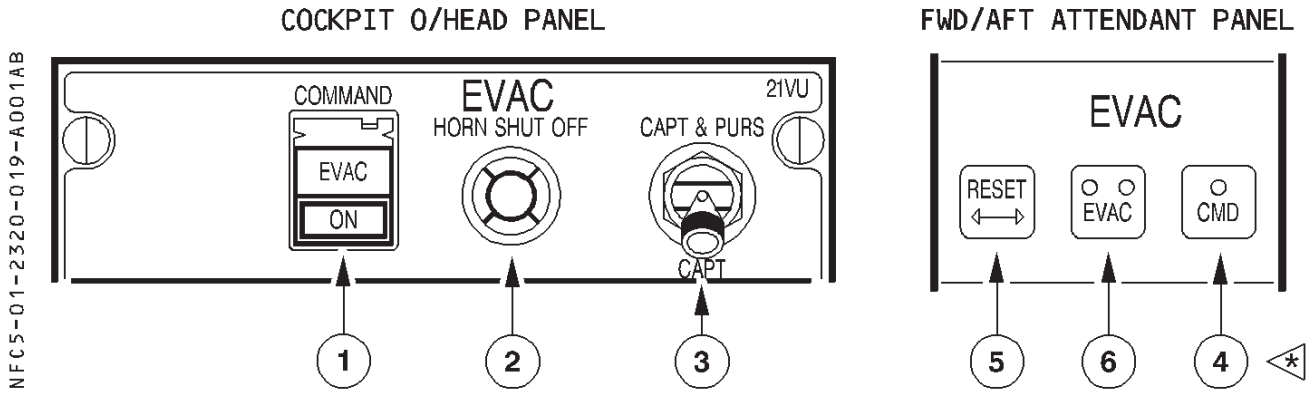
PA from cockpit

R

	PA TRANSMISSION KEY ON ACP	PA RECEPTION KNOB ON ACP	PUSH TO TALK ON HANDMIKE	PUSH TO TALK ON HANDSET
BOOMSET OR OXYGEN MASK	PRESSED	OUT		
HANDMIKE	PRESSED	OUT	PRESSED	
HANDSET				PRESSED

EMER EVAC



① COMMAND pb

ON : In the cockpit : – EVAC light flashes red.
– Horn sounds.
In the cabin : – EVAC lights flash at FWD and AFT attendant panels.
– Evacuation tone sounds.
Off : The alert is stopped.
The EVAC light flashes red when the alert is activated.

② HORN SHUT OFF pb

Pressing this button silences the cockpit horn.

③ CAPT and PURS/CAPT sw

CAPT and PURS : The alert may either be activated from the cockpit or the cabin.
CAPT : The alert may only be activated from the cockpit.
R If one of the cabin CMD pushbuttons is pressed, only the cockpit
R horn sounds for 3 seconds.

④ CMD pb

Pressing this button activates the alert, if the cockpit switch is at the CAPT & PURS position.
Pressing it again stops the alert.

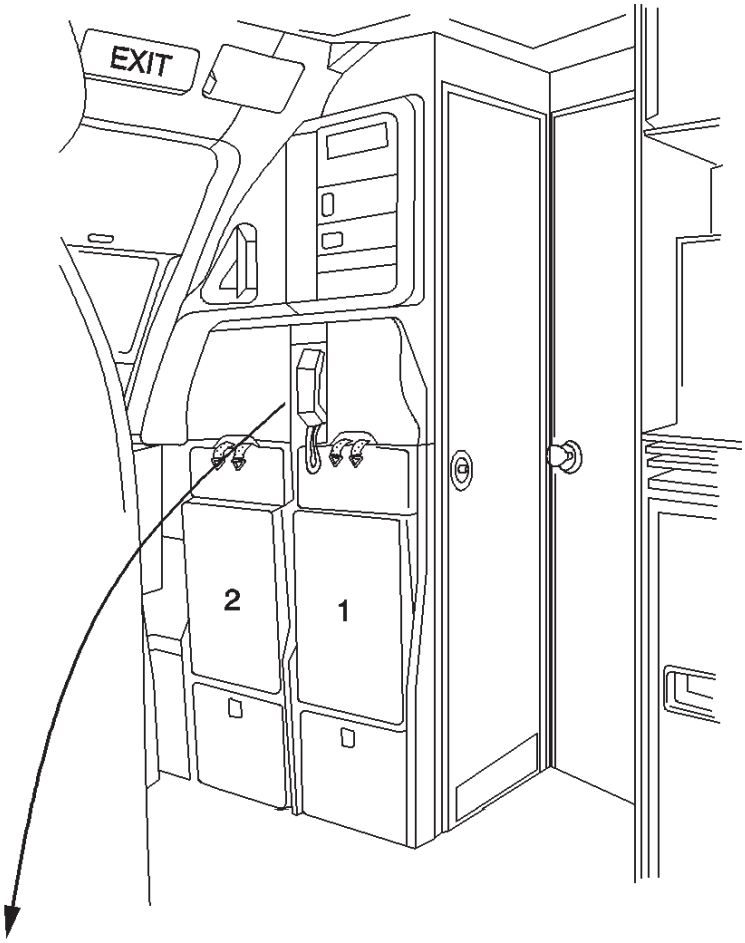
⑤ RESET pb

Pressing this button silences the EVAC tone.

⑥ EVAC lt

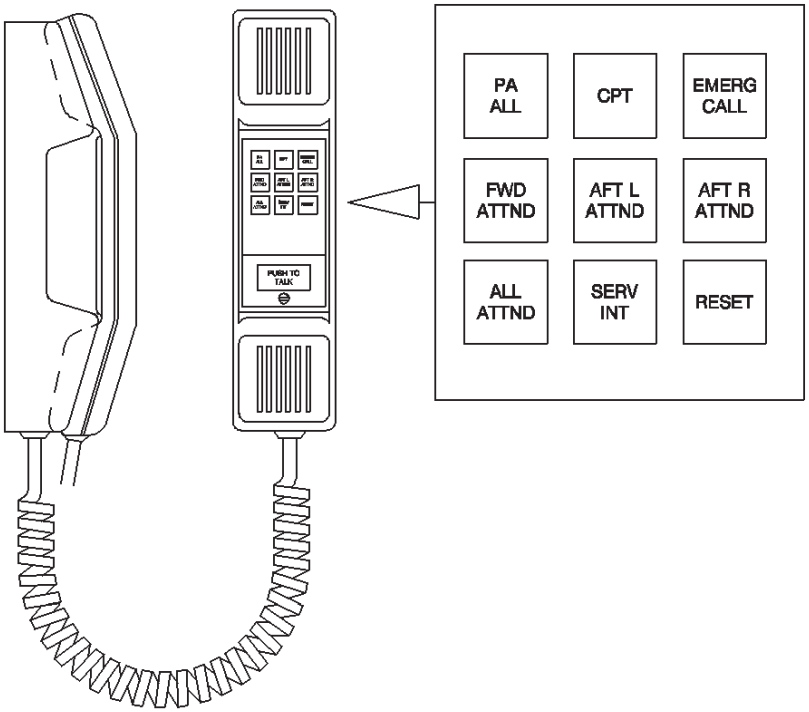
This light flashes when the alert is activated.

PURSER STATION



CA HANDSET

KEYBOARD



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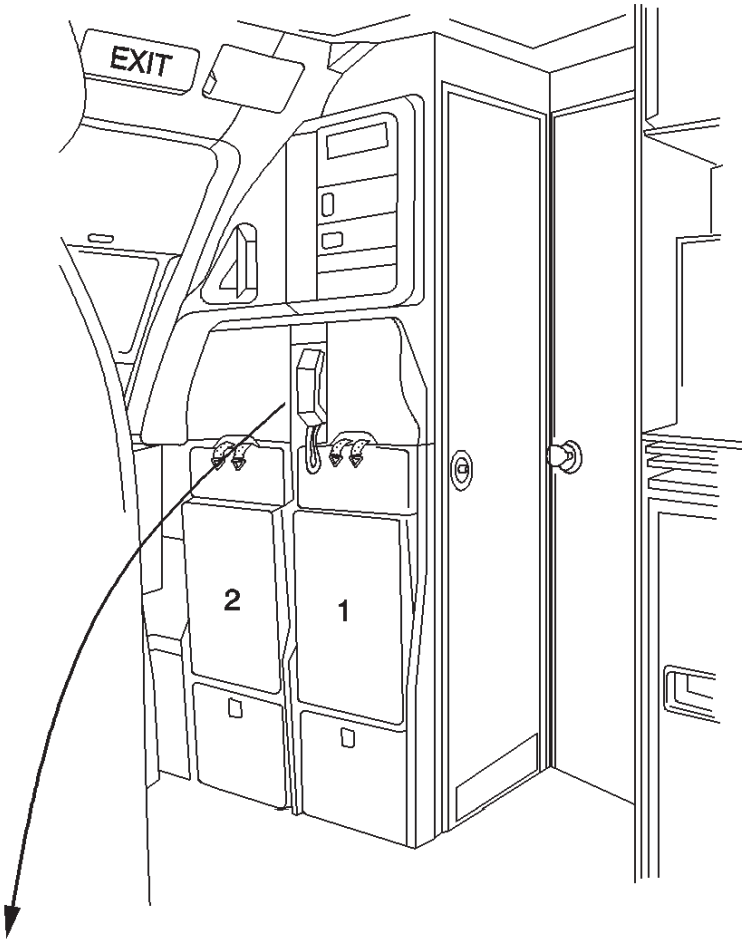
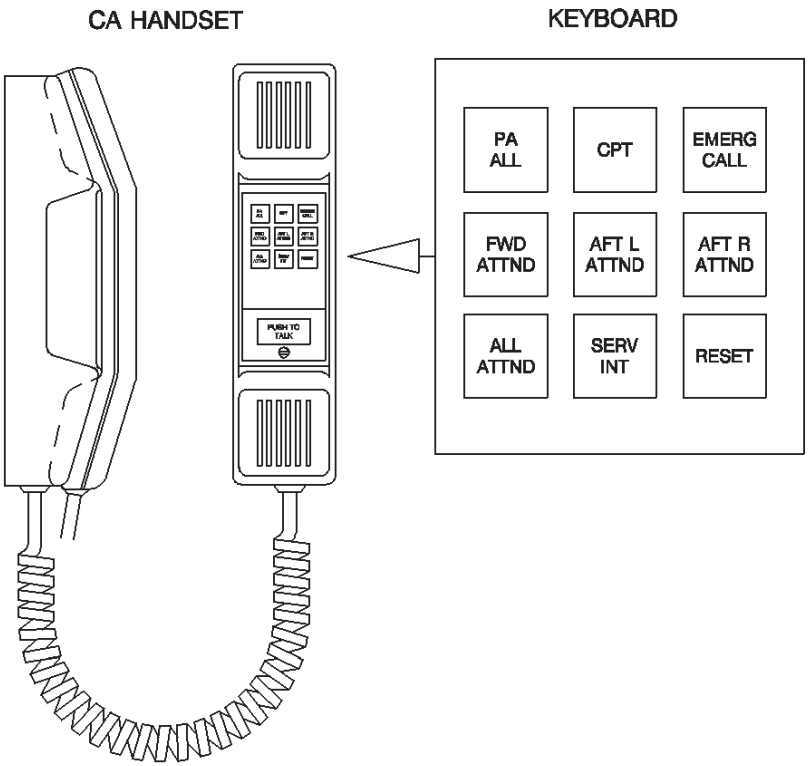
EMER EVAC

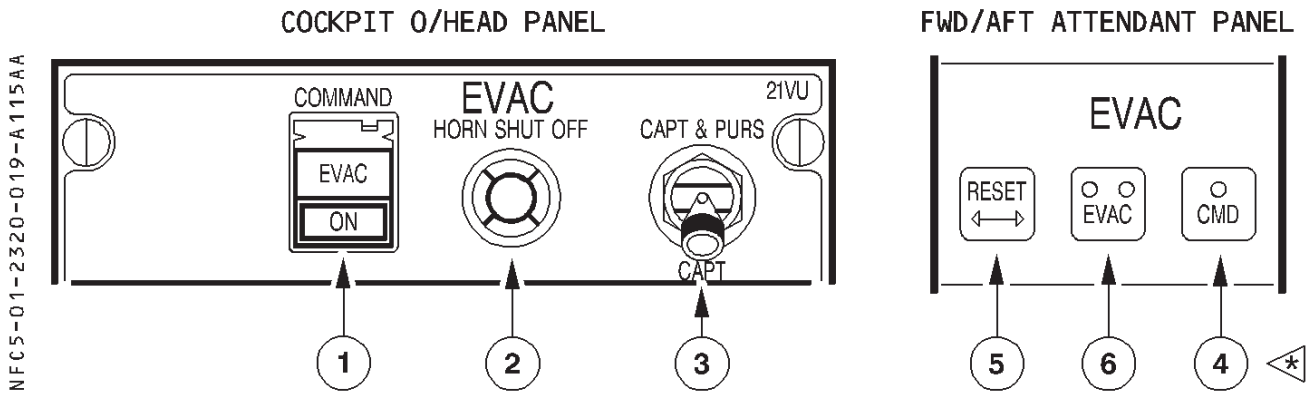
INTENTIONALLY LEFT BLANK

PURSER STATION





EMER EVAC



① COMMAND pushbutton

ON : In the cockpit : – EVAC light flashes red.
In the cabin : – EVAC lights flash at FWD and AFT attendant panels.
– Evacuation tone sounds.
Off : The alert is stopped.
The EVAC light flashes red when the alert is activated.

② HORN SHUT OFF pushbutton

Pressing this button silences the cockpit horn.

③ CAPT and PURS/CAPT switches

CAPT and PURS : The alert may either be activated from the cockpit or the cabin.
CAPT : The alert may only be activated from the cockpit.
If one of the cabin CMD pushbuttons is pressed, only the cockpit horn sounds for 3 seconds.

④ CMD pushbutton

Pressing this button activates the alert, if the cockpit switch is at the CAPT & PURS position.
Pressing it again stops the alert.

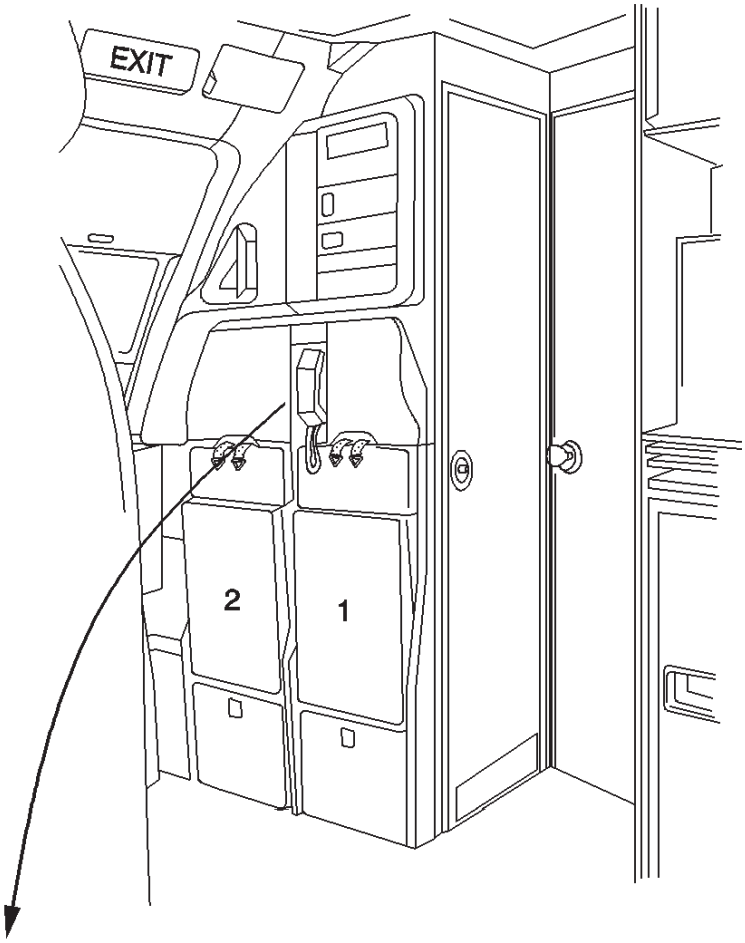
⑤ RESET pushbutton

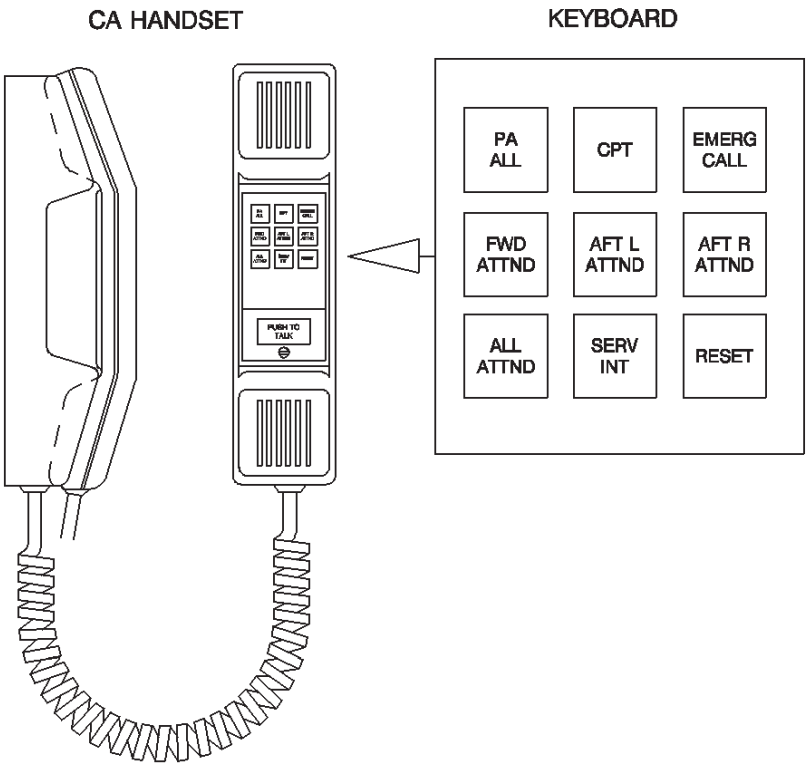
Pressing this button silences the EVAC tone.

⑥ EVAC lt

This light flashes when the alert is activated.

PURSER STATION





NFC5-01-2320-020-A002AA

DESCRIPTION

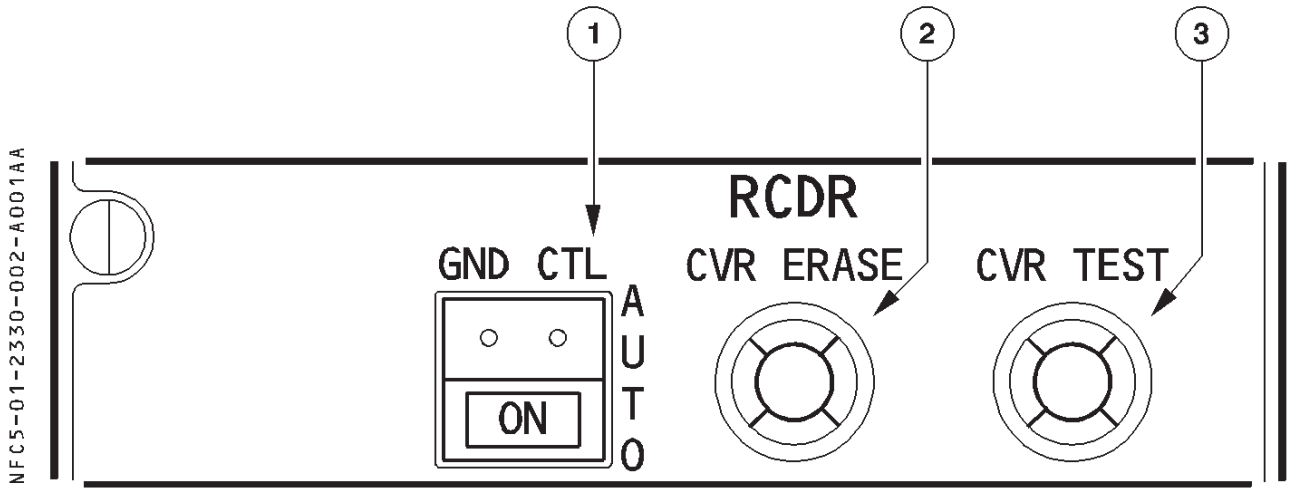
The cockpit voice recorder (CVR) records :
- direct conversations between crew members in the cockpit
- all aural warnings sounded in the cockpit
- communications received and transmitted by radio
- intercommunication conversations between crew members
- announcements transmitted over the passenger address system, if PA reception is selected on the third audio control panel.

Only the last 30 minutes of recording are retained.
The CVR system consists of :
- a remote microphone behind the overhead panel.
- a “hot mike” function, which records the crew members voice directly from their microphone, even if the push to talk switch is not activated.
- a crashproof four-track recorder, equipped with an underwater locating beacon, in the aft section of the aircraft
- a control panel on the overhead panel.

It is energized automatically :
- on the ground during the first 5 minutes after the aircraft electrical network is energized
- on the ground with one engine running
- in flight

On the ground, it is stopped automatically 5 minutes after the last engine shutdown.
On the ground, the crew can energize the CVR manually by pressing the GND CTL pushbutton.

CONTROLS AND INDICATORS ON OVERHEAD PANEL



(1) GND CTL switch (spring-loaded)

ON	: The CVR, DFDR, and QAR ◁ are on. The ON light comes on blue.
AUTO	: The CVR, DFDR, and QAR ◁ are on, according to the logic. (See page 1).

(2) CVR ERASE pushbutton

Pressed for 2 seconds	: This completely erases the tape, if : – The aircraft is on the ground, and – The parking brake is on.

(3) CVR TEST pushbutton

Pressed and held	: This activates the test, if the CVR is on (the GND CTL pushbutton pressed, or during the first 5 minutes after energization of the aircraft electrical network), and the parking brake is on.

R Refer to the FCOM 3.03.06, p. 2 for additional
R information.

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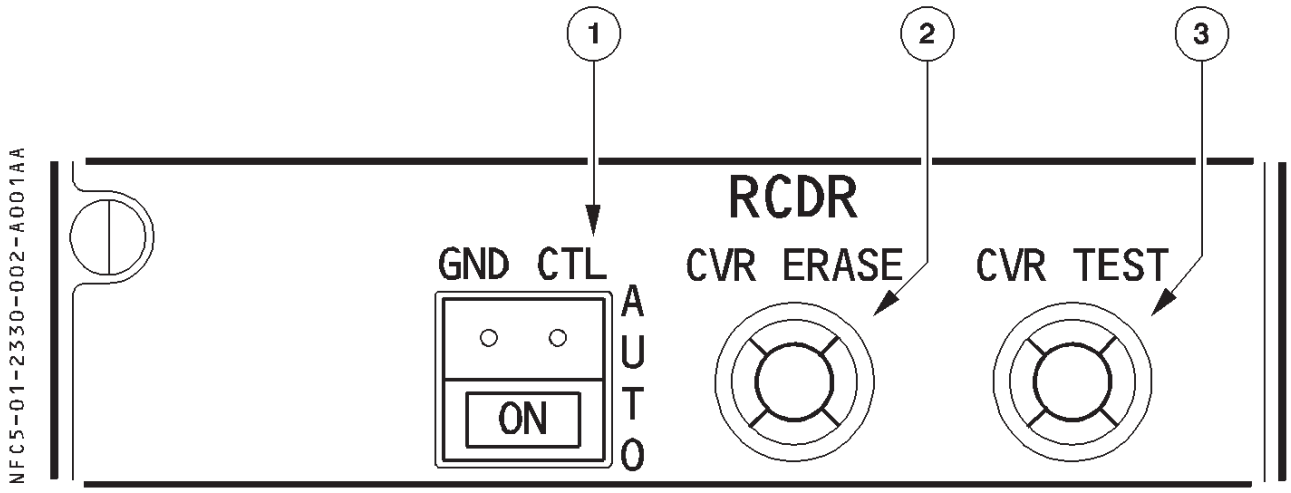

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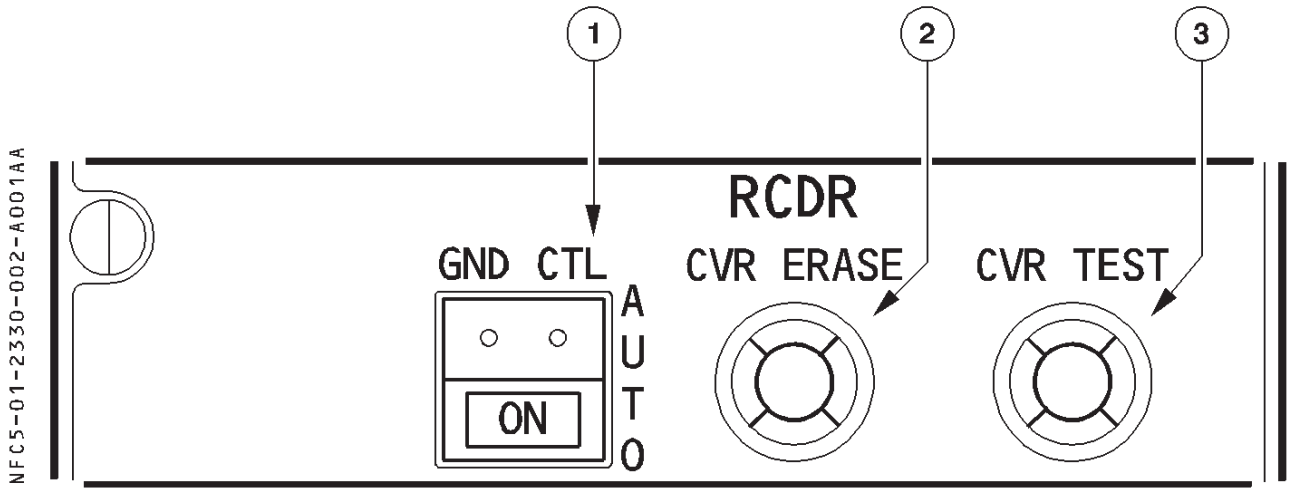

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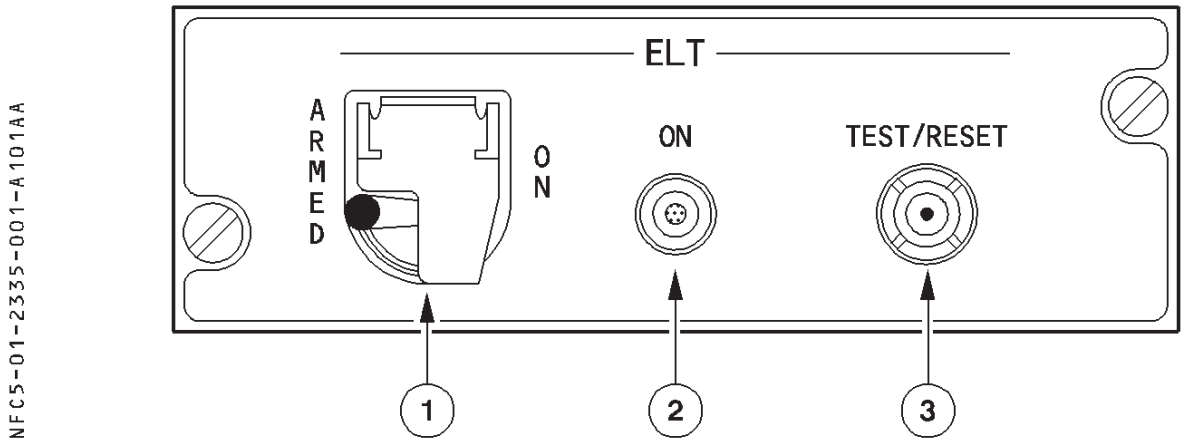
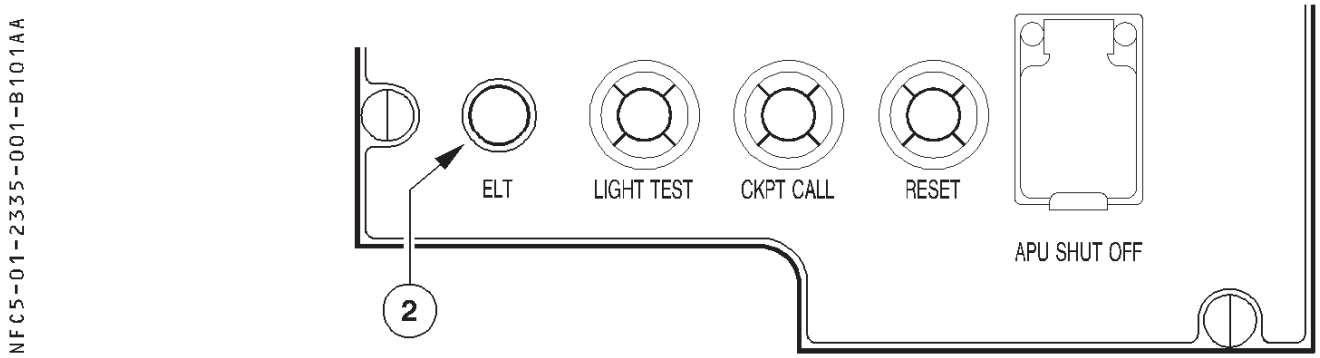
CONTROLS AND INDICATORS

OVERHEAD PANEL



EXTERNAL POWER PANEL



(1) ELT sw

The guard keeps this switch in the ARMED position.
ARMED : In case of impact, the Emergency Locator Transmitter (ELT) transmits an emergency signal (on 121.5, 243 and 406.025 MHZ).
ON : The ELT transmits an emergency signal.

(2) ON and ELT lights

These lights comes on amber when the emergency signal is transmitted or during ELT autotest.

(3) TEST/RESET pb

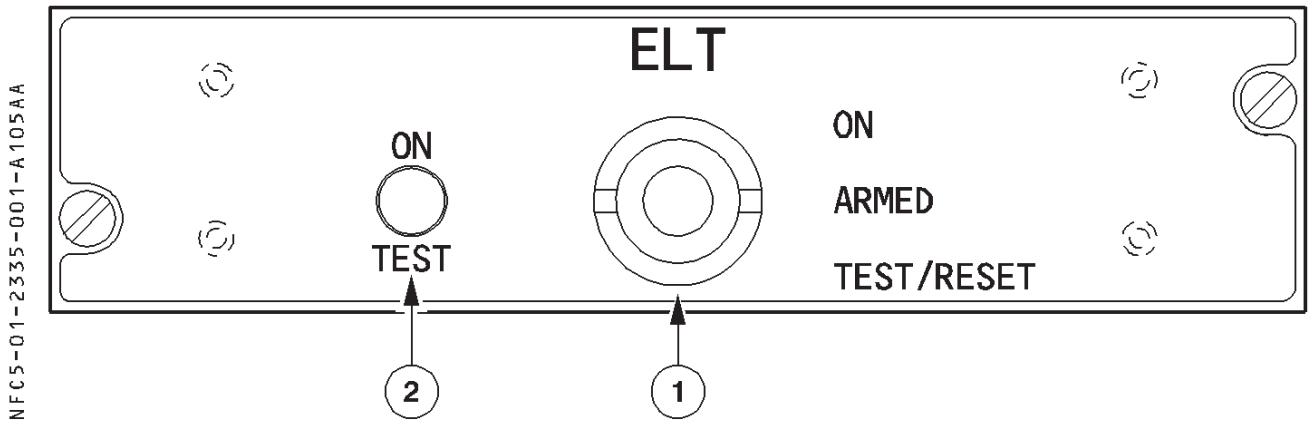
Pressing this pushbutton starts the ELT auto test.

Note : If the ELT is triggered, by an impact in ARMED mode, pressing this pushbutton resets the ELT and stops the signal transmission.

CONTROLS AND INDICATORS

OVERHEAD PANEL

R



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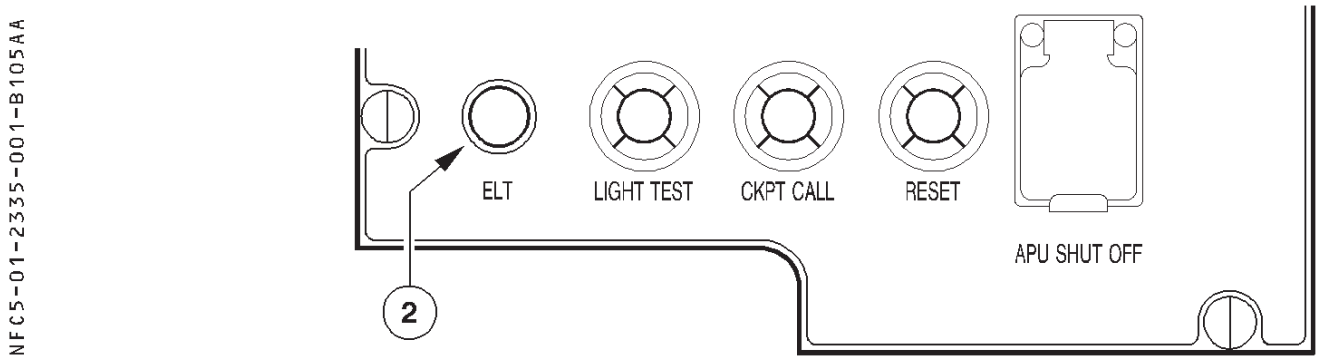


(1) ELT sw

R

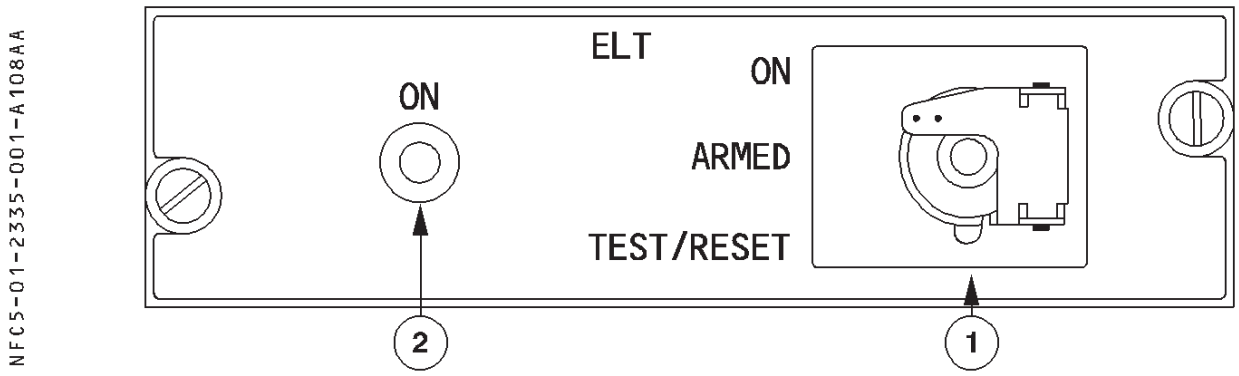
ON	:	The Emergency Locator Transmitter (ELT) transmits an emergency signal.
ARMED	:	In case of impact, the ELT transmits an emergency signal (on 121.5, 243 and 406.025 MHZ).
TEST/RESET	:	Starts the ELT autotest.

Note : If the ELT is triggered by an impact in ARMED mode, selecting TEST/RESET resets the ELT and stops the signal transmission.

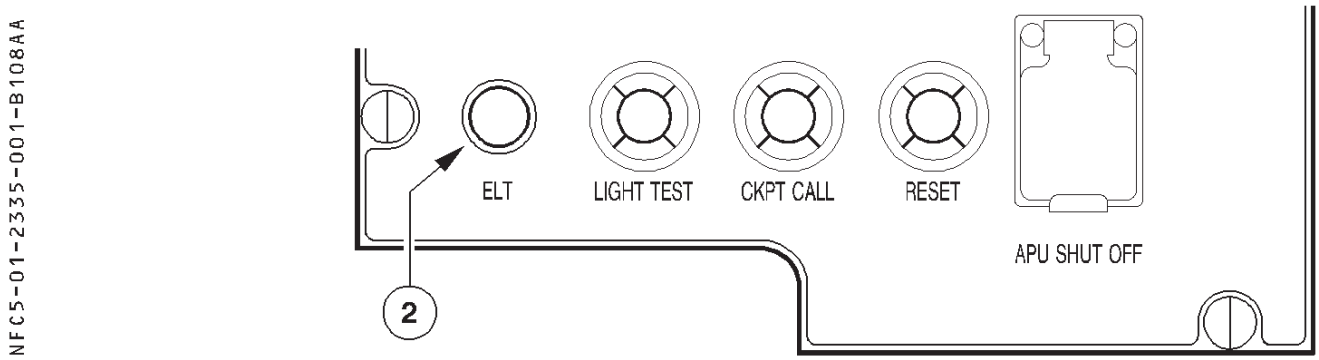
(2) ON TEST and ELT lights

These lights come on amber, either when the emergency signal is transmitted or during the ELT autotest.

R CONTROLS AND INDICATORS



EXTERNAL POWER PANEL



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ARMED : In case of impact, the ELT transmits an emergency signal (on 121.5, 243, and 406 MHz).

TEST/RESET : Starts the ELT autotest.

Note : If the ELT is unduly triggered in ARMED mode (by an external impact, hard landing, etc), select the TEST/RESET position to reset the ELT and stop signal transmission.

(2) ON and ELT lights

These lights come on amber when the emergency signal is transmitted or during the ELT autotest.

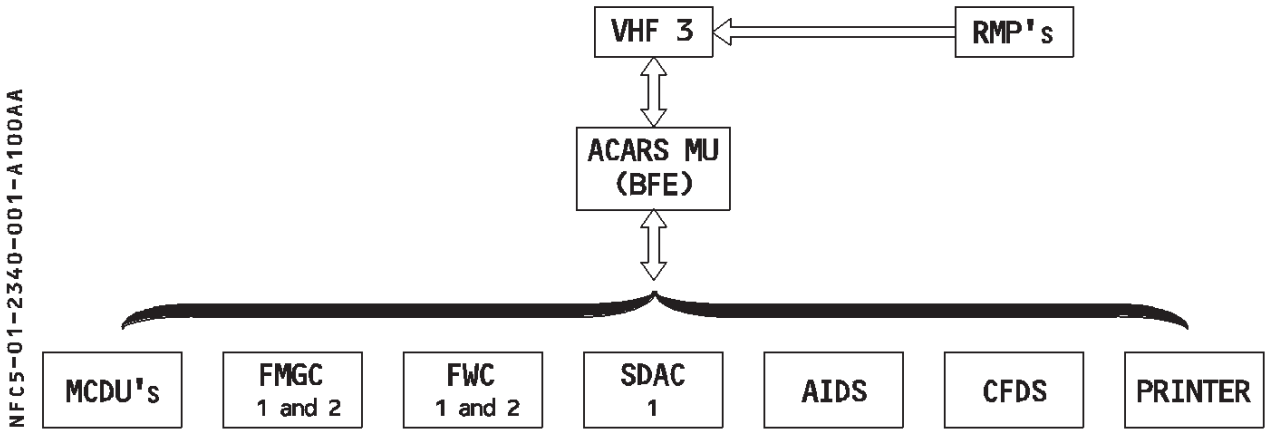
GENERAL

INTRODUCTION

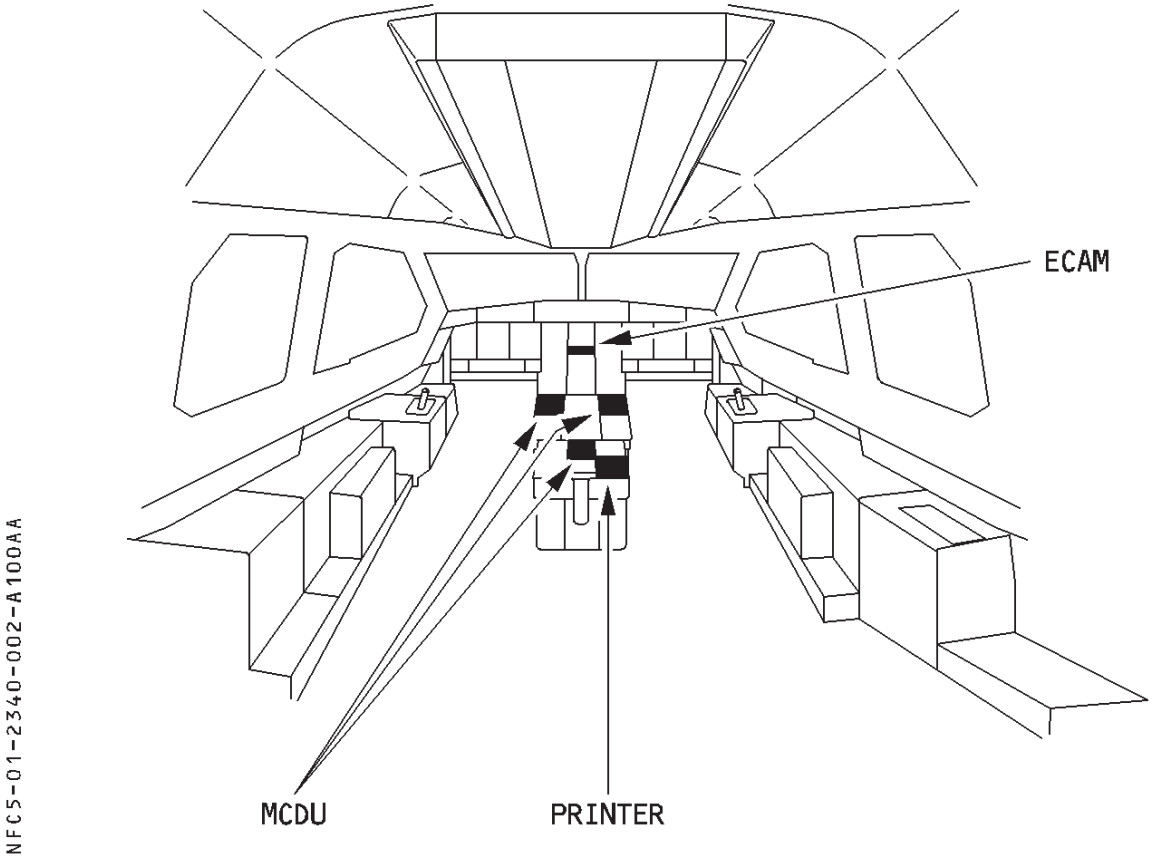
The Aircraft Communication Adressing and Reporting System (ACARS) allows direct exchange of data between aircraft and airline ground computer through the VHF 3.
Aircraft to ground messages (downlink) comprise operational, maintenance, monitoring, performance and cabin data.
Ground to aircraft messages (uplink) contain crew information (wind for example) or might request for transmission of data which are sent automatically or by crew action.
Automatic downlink of reports is carried out by ACARS Management Unit [MU), which is programmed according to airline needs (Buyer Furnishing Equipment).
Due to the highly customized programming, the ACARS functions may vary for different airlines and therefore are not described in detail.

SYSTEM ARCHITECTURE

The ACARS system consists of a Management Unit (MU) connected to the following elements :



COCKPIT ARRANGEMENT



ACARS operation is performed through the already available cockpit equipment:
- ECAM for operational indications.
- MCDU for control of ACARS related functions.
- PRINTER for hard-copies.

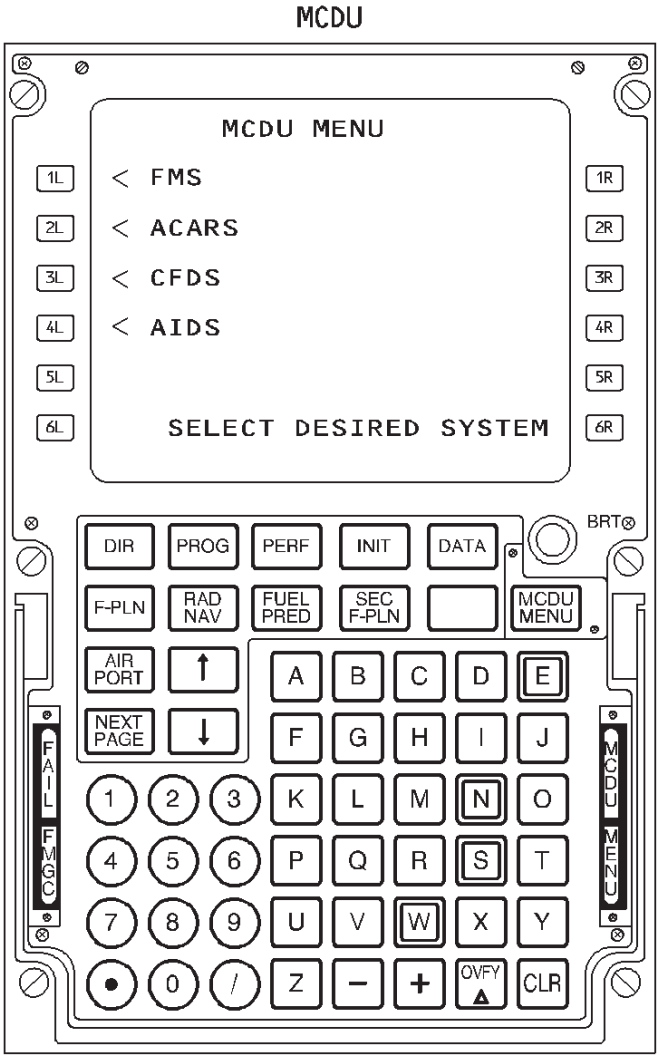
R The connector for portable data loader is in the avionics bay.

OPERATION MODES

MANUAL ACCESS TO ACARS FUNCTIONS

ACARS functions are manually selected through MCDU. They are obtained for FMGC, CFDS or AIDS by selecting the corresponding key. Cabin management functions (if available) are accessible through the cabin installed system.

R



(NOTE : DEPENDING ON OPTIONS, MCDU FRONT FACE MAY SLIGHTLY DIFFER)

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Pressing of a system key will display the system pages as defined in the airline customized MU programming.

AUTOMATIC ACCESS TO ACARS FUNCTIONS

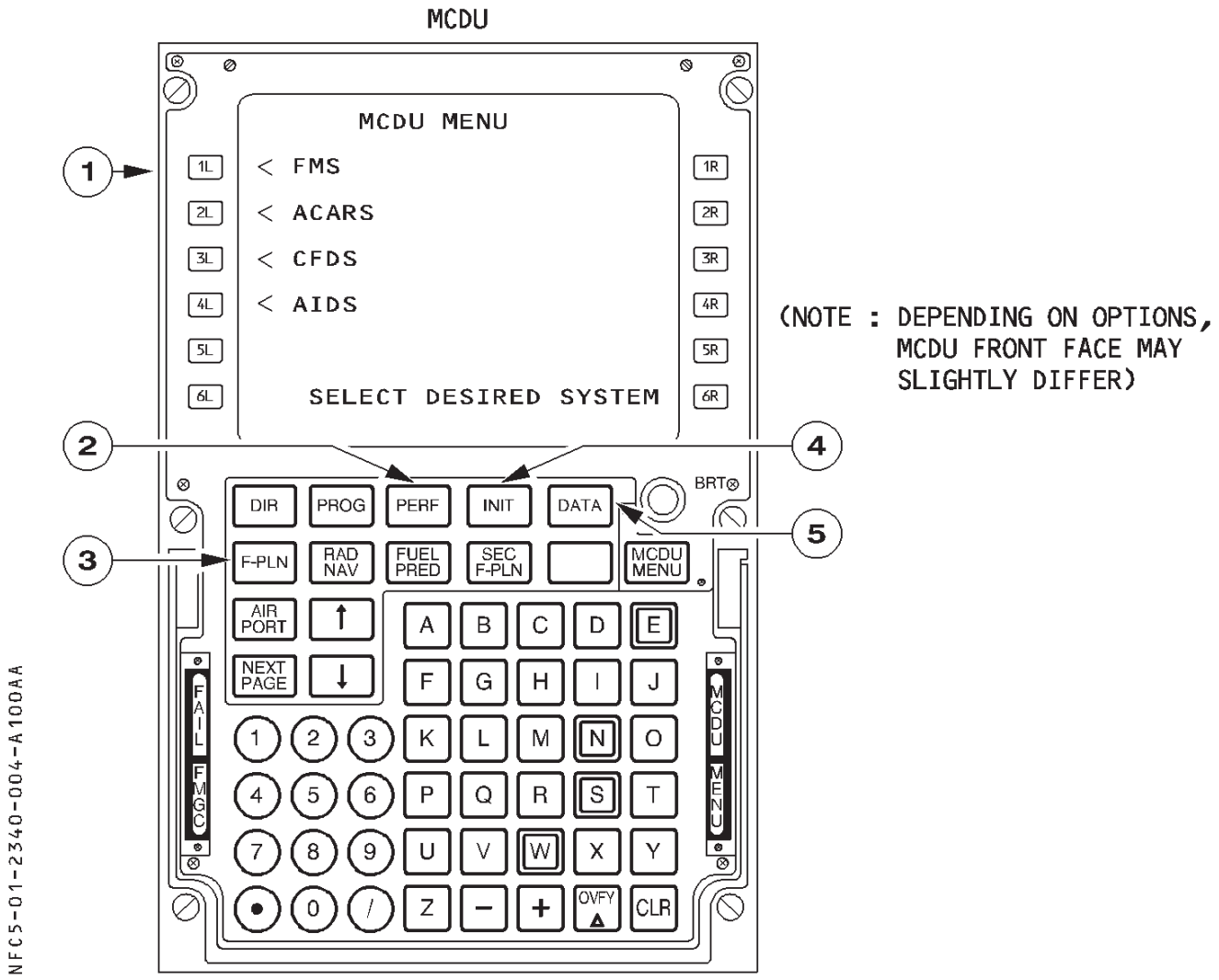
Automatic data transmission can be initiated by MU or aircraft systems programming or an uplink message. There is no cockpit indication nor crew action is required. It is a dialogue between ground and aircraft computers.

ACARS FUNCTIONS

Data and reports of a particular system are available through Airbus defined interfaces (seller furnishing equipment).

FMS

R



(1) Pressing key selects related system then

(2) Pressing key gives access to TO data (Uplink only)

(3) Pressing key gives access to wind data (F-PLN page)

(4) Pressing key gives access to F-PLN initialisation and wind data (Uplink only)

(5) Pressing key gives access to Pre-flight, Post-flight report and ACARS print/program (downlink only).
For operation see FMGS PILOT GUIDE (Refer to 4.04.40).

ACARS

Pressing ACARS key displays airline defined functions (message transmission , event times for example).

CFDS (Refer to 1.45.20)

The CMS ACARS interface permits to downlink the following data :
- Post flight report (on ground) or current flight report (in flight) which concerns :
  - all failure messages detected by the BITEs
  - the warnings displayed to the crew during the last or current flight leg.

  Reports can be downlinked upon crew or ACARS MU request.
- Avionics data which concerns the individual system BITE data (manual downlink only).
- Failure messages and warnings (broadcast data transmitted in real-time to the MU).
- Class 3 report (on ground) containing all class 3 failures detected during the last flight leg. The report can be downlinked upon crew or ACARS MU request.

AIDS

The AIDS ACARS interface transmits to the ACARS MU the data for the following applications:
- Aircraft Performance Monitoring (APM),
- Engine Condition Monitoring (ECM),
- APU Health Monitoring (AHM).

Any of the AIDS DMU reports can be downlinked (transmitted to the MU):
- manual on the ground or in flight upon crew request,
- automatically in real-time,
- upon ACARS MU (ground or automatic) request.

AUTOMATIC DOWNLINK OF REPORTS

Automatic downlink of reports are provided by ACARS. Each report generated by a peripheral system may be downlinked depending on each airline MU programming.

UPLINK MESSAGES

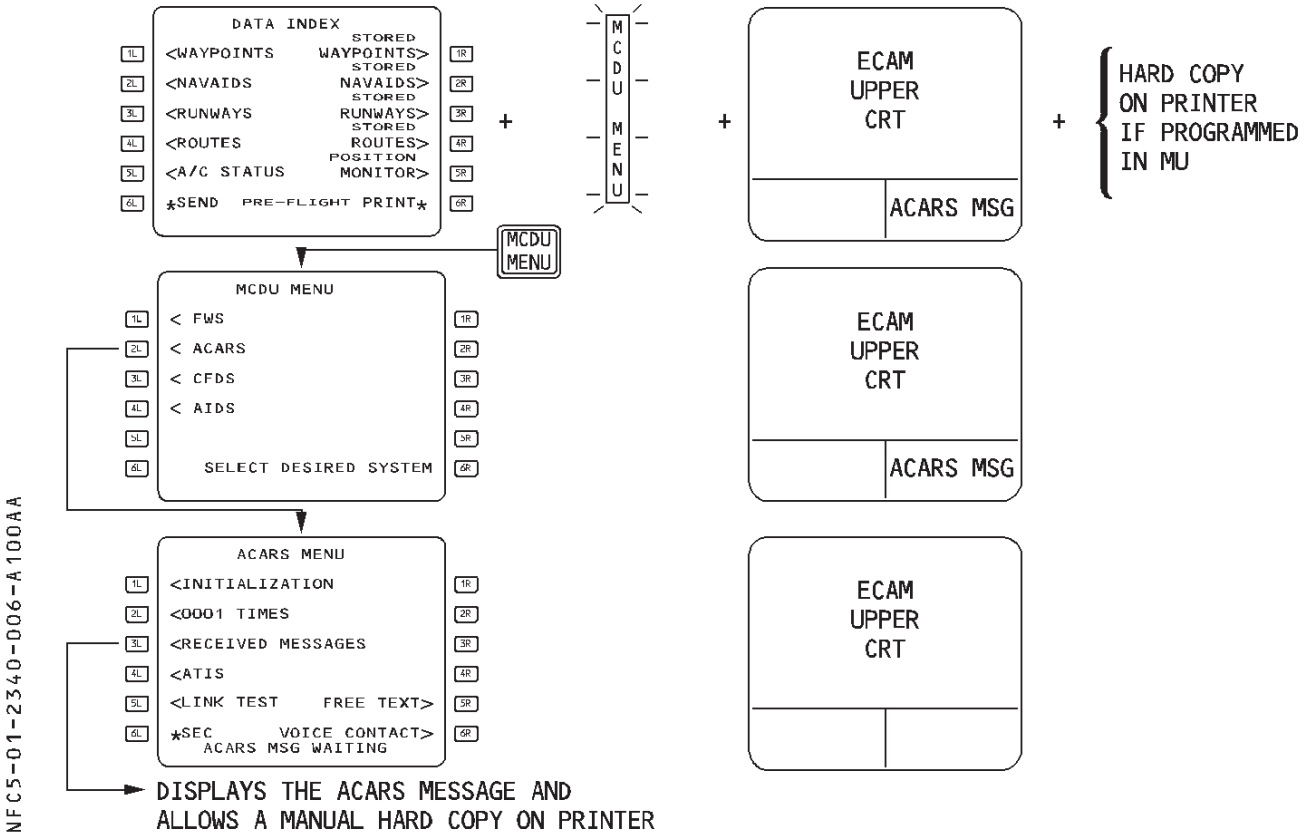
Two types of uplink messages are provided :

· Messages not indicated to the crew :
There is no indication given to the crew, neither for uplink message nor for downlink answer. It is a dialogue between ground and aircraft computers.

· Messages indicated to the crew by :
– "ACARS MSG" advisory (pulsing green) on ECAM Memo

Note : A steady green "ACARS STBY" advisory is displayed in case of ACARS communications loss between aircraft and ground.

– Message on MCDU (ACARS MSG WAITING for example) or MCDU MENU light illumination if the MCDU is not in the mode where uplink message can be displayed in order to select the correct mode (FMGC, ACARS, AIDS, CFDS)
– Hard copy on cockpit printer depending on MU programming.
Example of uplink message indication with MCDU in FMS mode :



VOICE/ACARS transfer on VHF 3

VHF 3 can be used in VOICE mode in case of :

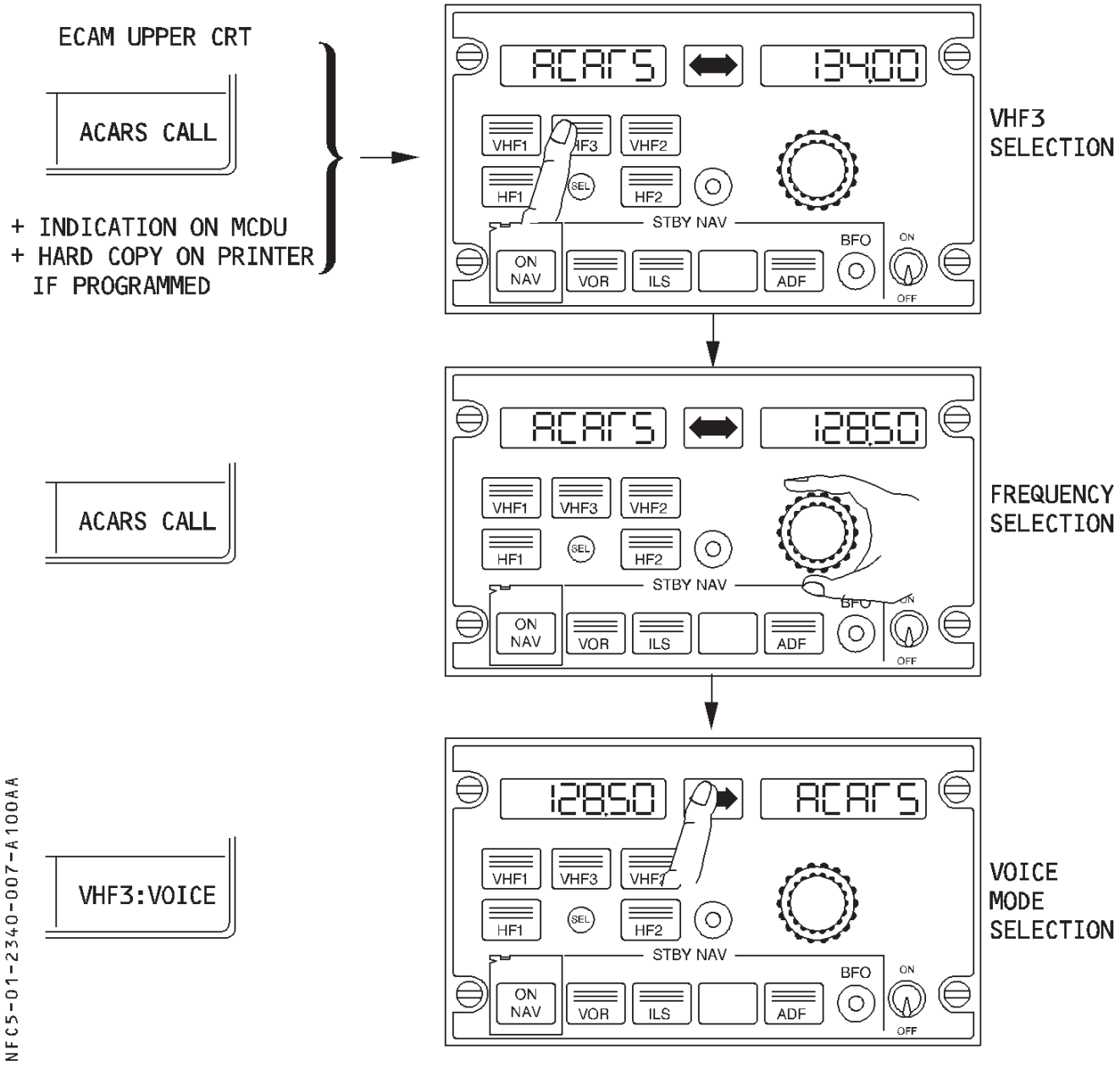
- VHF 1 or VHF 2 failure
- ACARS CALL

  The green pulsing "ACARS CALL" advisory indicates that a message requesting a voice conversation has been received from the ground.

  The green "VHF 3 : VOICE" advisory indicates that the VHF 3 transceiver operates in voice mode, therefore the ACARS communications are interrupted.

  Depending on MU programming, the ACARS frequency may be either automatically tuned by the ACARS MU or manually tuned through the RMP.

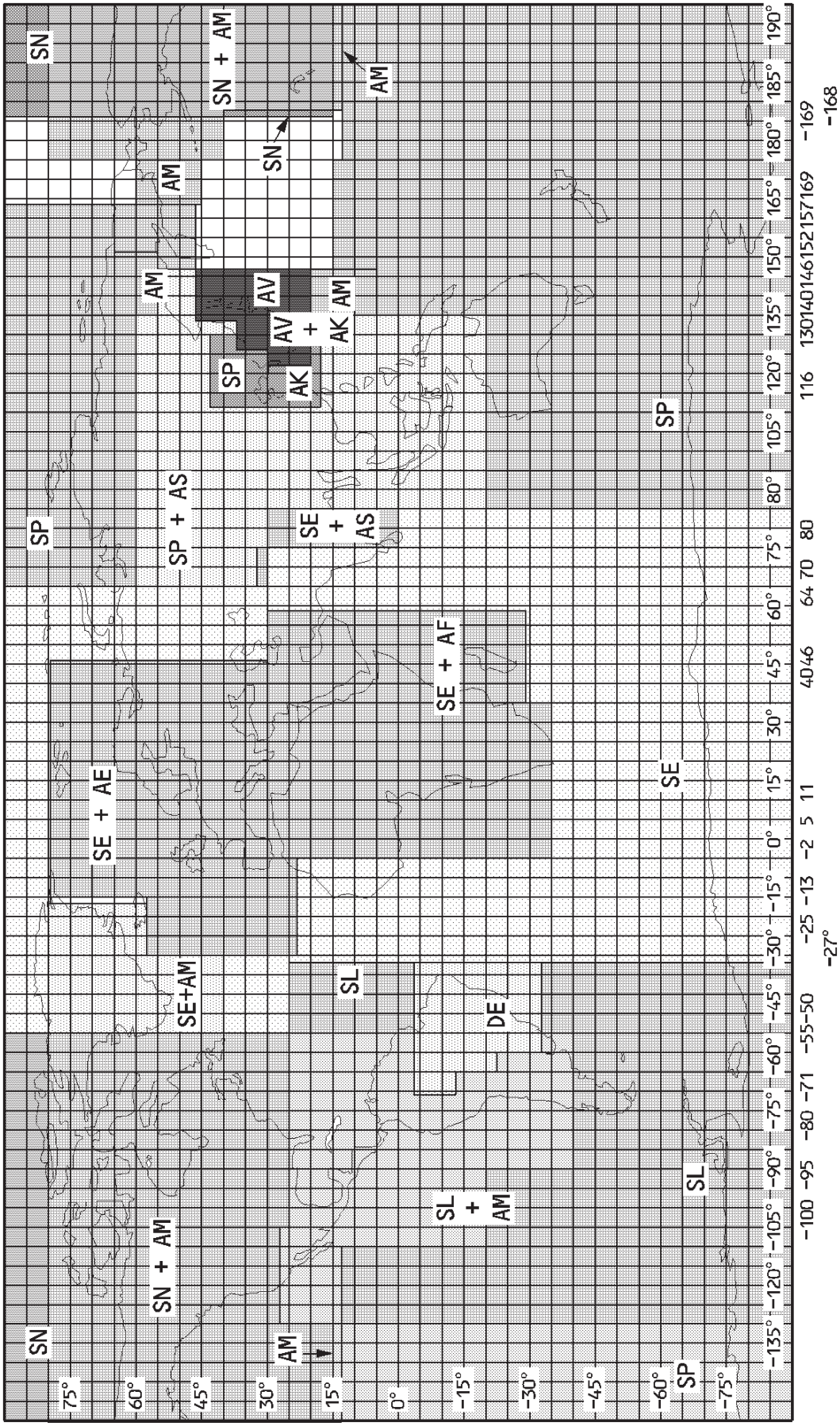
  In this case, the ACARS/VOICE transfer is made directly on any of RMPs (without using MCDU).



Return to ACARS mode is provided by pressing again the transfer key on RMP.

WORLD MAP ACARS FREQUENCIES

R



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WORLD MAP ACARS FREQUENCIES (CONT'D)

The table below defines the world zone abbreviations, indicates their associated Service Provider, MCDU label, and ACARS frequency.

ABBREVIATION	SERVICE PROVIDER	MCDU LABEL	FREQUENCY	Family
SP	SITA PACIFIC	SIT-PAC	131.550 MHz	SITA
SN	SITA NORTH AMERICA	SIT-NAM	136.850 MHz	SITA
SL	SITA LATIN AMERICA	SIT-LAM	131.725 MHz	SITA
SE	SITA EUROPE	SIT-E/A	131.725 MHz	SITA
DE	DEPV BRAZIL	DEPV	131.550 MHz	SITA
AV	AVICOM	AVICOM	131.450 MHz	SITA
AM	ARINC AMERICA	ARI-AM	131.550 MHz	ARINC
AE	ARINC EUROPE	ARI-EUR	136.925 MHz	ARINC
AF	ARINC AFRICA	ARI-AFR	126.900 MHz	ARINC
AK	ARINC KOREA	ARI-KOR	131.725 MHz	ARINC
AS	ARINC ASIA	ARI-ASI	131.450 MHz	ARINC

GENERAL

The Satellite Communication (SATCOM) system allows the exchange of information between the aircraft and a Ground Earth Station (GES), via geosynchroneous satellites.
It provides up to six independent channels :
One channel is used for data transmissions (ATSU or ACARS). Two, three or five channels are used for voice transmissions (cockpit or cabin voice). The cockpit voice function must be activated, in order to be available. The cabin telephone system must be installed, to be able to use cabin voice function.
The ATSU or ACARS normally transmit via VHF3. They automatically switch to SATCOM when VHF3 is not available.
The cockpit voice interface is controlled by the Audio Control Panels (ACPs) for call set-up and call termination, and by the MCDU for the call number selection. It allows the crew :
– To initiate air to ground calls and to receive ground to air calls.
– To select the call priority, in case of air to ground calls.
– To use manual dial or pre-recorded phone numbers.
If a SATCOM telephone handset is installed in the cockpit, the crew can set up a call without the use of the MCDU.
SATCOM functions are programmed through the Owner Requirement Table (ORT), according to airline needs.
Due to the highly customized programming, the SATCOM functions may vary for different airlines and are, therefore, not described in detail.

CONTROLS AND INDICATORS

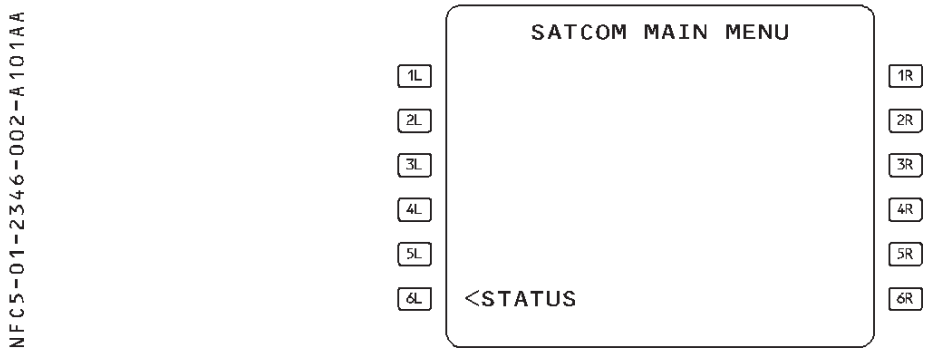
ACP INTERFACE

Refer to 1.23.20 description.

MCDU INTERFACE

SATCOM MAIN MENU PAGE

The crew accesses this page by selecting SAT on the MCDU MAIN page.



6L This key provides access to the SATCOM STATUS page, which contains LOG ON and channel status information.

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CONTROLS AND INDICATORS

ACP INTERFACE

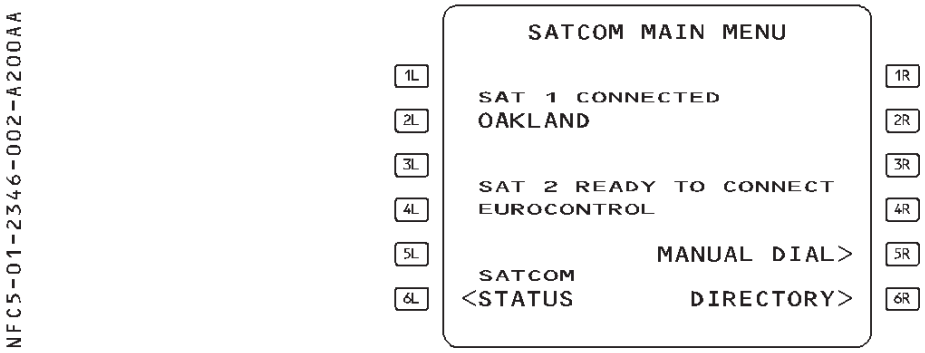
Refer to 1.23.20 description.

MCDU INTERFACE

SATCOM MAIN MENU PAGE

The crew accesses this page by selecting SAT on the MCDU MAIN page.

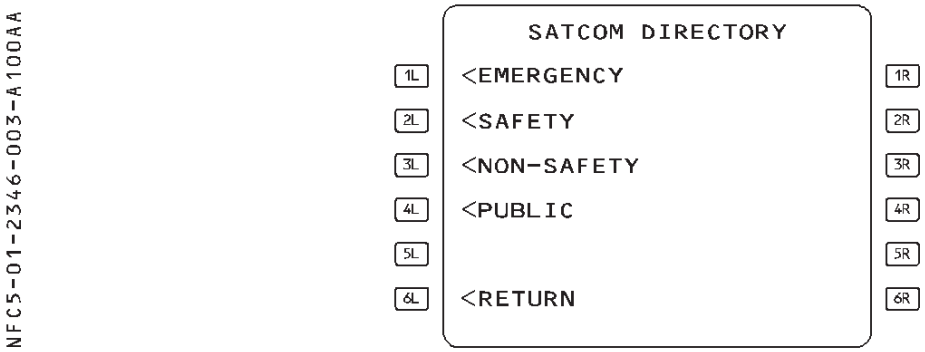
R



R 2L (4L) (Label line) : This field displays the SATCOM channel 1 (2) status :
R – READY TO CONNECT : The channel is ready to support a call.
R – NOT AVAILABLE : The channel is not available (failed or not logged).
R – DIALING : Cockpit call in progress.
R – INCOMING CALL : Advises of an incoming ground to air call.
R – CONNECTED : The circuit is connected.
R – CALL FAILED : The transmission is interrupted.
R 2L (4L) (Data line) : This field displays :
R – The title of the selected phone number, in case of an air to ground call.
R – The number, if the MANUAL DIAL option is used.
R – GRND-AIR CALL, in case of a ground to air call.
R 5R : This key provides access to the Manual Dial page. This page allows the dialing of a phone number.
R 6L : This key provides access to the SATCOM STATUS page, which contains LOG ON and channel status information.
R 6R : This key provides access to the SATCOM DIRECTORY PAGE.

SATCOM DIRECTORY PAGE

R This page provides access to 4 phone number lists, where phone numbers can be
R memorized, according to their priority.
R



R 1L : EMERGENCY for Priority 1 - Reserved for emergency and distress phone
R numbers only.
R 2L : SAFETY for Priority 2 - Reserved for regulatory and flight safety phone
R numbers only.
R 3L : NON-SAFETY for Priority 3 - Reserved for non flight safety phone numbers.
R 4L : PUBLIC for Priority 4 - Reserved for personal phone numbers.
R 6L : This key is used to return to the SATCOM MAIN MENU page.

SATCOM CATEGORY NUMBERS PAGE

R The CATEGORY NUMBERS page provides access to the pre-recorded phone numbers.
R As an example, the following figure shows the SAFETY CATEGORY NUMBER page.
R

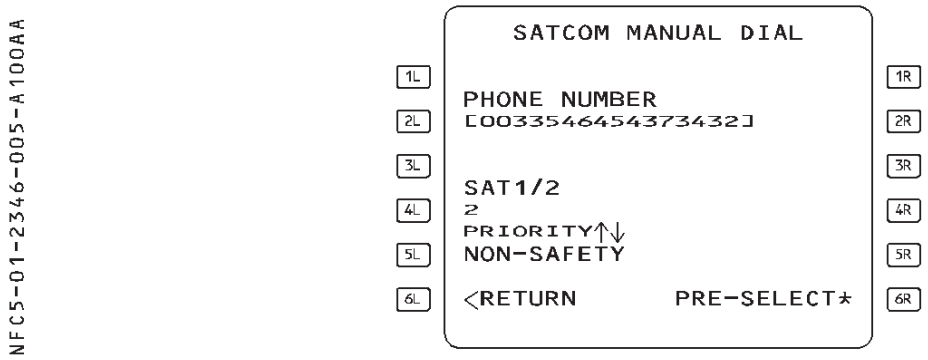
NFC5-01-2346-004-A100AA

```
        SATCOM SAFETY   1/3
                       SAT1/2
1L  *IOR CONTROL            1   1R
     00495218796214
2L  *CDG ARPT                   2R
     [00985647213369 85]
3L  *HGK ARPT                   3R
     004632189752123
4L  *ORD ARPT           SORT*   4R
     453356722268
5L  *ORY ARPT           FIND*   5R
     [004433566213]     [    ]
6L  <RETURN                     6R
```

R 1L, 2L, 3L, 4L, 5L : These fields display the phone numbers and their titles.
R When pressed, they dial the corresponding phone number.
R There are two types of numbers :
R - Protected : Displayed in green.
R - Unprotected : Displayed in blue brackets.
R 1R : This field displays the selected SATCOM channel.
R 4R : This function alphabetically sorts the phone numbers, within the
R category, by title.
R 5R : This function automatically searches for a phone number from
R the beginning of this category, by entering up to the first three
R letters of the title into the scratchpad, and by pressing 5R.
R 6L : This key is used to return to the SATCOM DIRECTORY page.

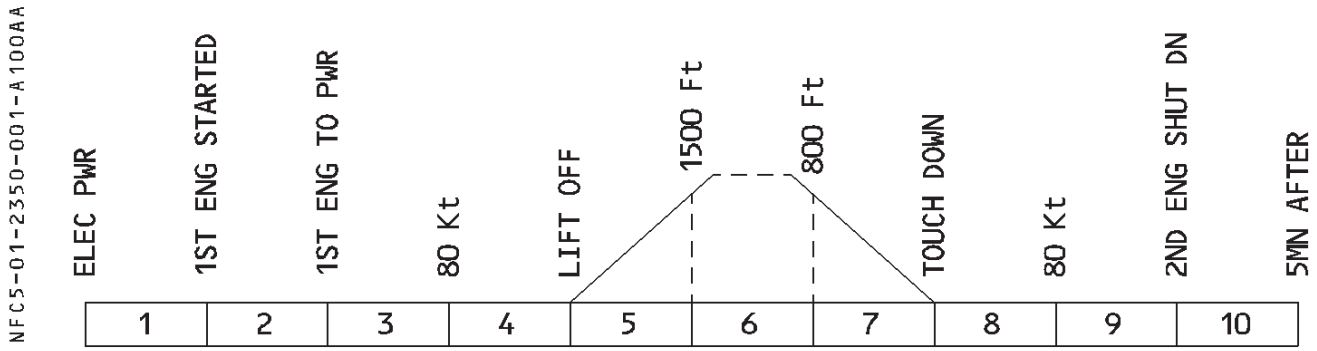
SATCOM MANUAL DIAL PAGE

The MANUAL DIAL PAGE enables the crew to initiate an air to ground call by manually entering a phone number.



2L (data line)	:	This field displays the phone number in blue brackets, after having been entered in the scratchpad.
4L	:	This field displays the selected SATCOM channel.
5L	:	This field displays the priority for the manual dial number. The priority can be changed by pressing the slew up or down keys on the MCDU keyboard.
6L	:	This key is used to return to the SATCOM MAIN MENU page.

WARNINGS AND CAUTIONS



E / WD: FAILURE TITLE conditions	AURAL WARNING	MASTER LIGHT	SD PAGE CALLED	LOCAL WARNING	FLT PHASE INHIB
CIDS 1 + 2 FAULT	SINGLE CHIME	MASTER CAUTION	NIL	NIL	3,4,5 7,8
VHF 1(2) (3) EMITTING HF 1 (2) EMITTING Transmitter emitting more than 60 s					
SATCOM FAULT (◁)					
ACARS FAULT (◁)	NIL	NIL			

MEMO DISPLAY

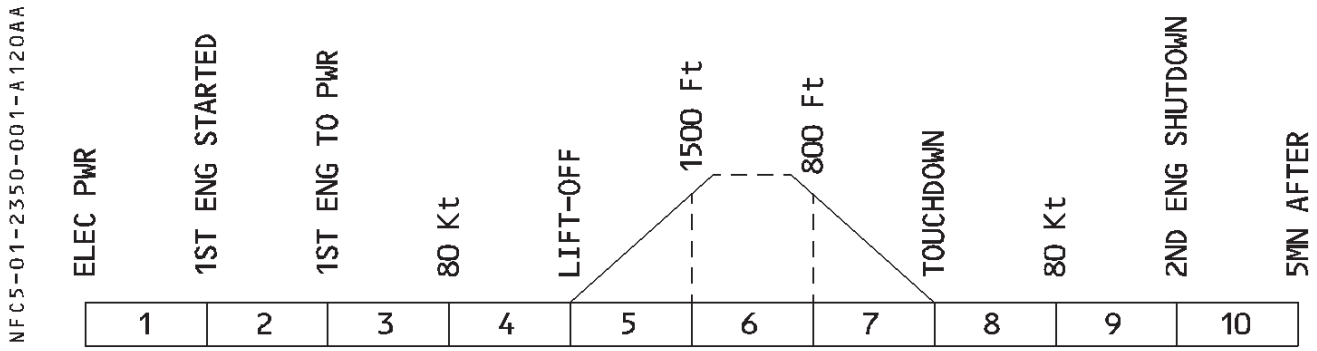
This display shows AUDIO 3 XFRD in green, if the AUDIO SWITCHING selector is not on NORM.

If ACARS is installed, the display shows :

- ACARS CALL in green, pulsing, if ACARS has received a message from the ground requesting a voice conversation
- VHF 3 : VOICE in green, if VHF3 is operating in voice mode and ACARS communication is interrupted
- ACARS MSG in green, pulsing, if ACARS has received a message from the ground
- ACARS STBY in green, if ACARS communications between the aircraft and the ground are lost

In addition, if SATCOM is installed the display shows SATCOM ALERT in green, when a message with a priority level below 4 is received from the ground.

WARNINGS AND CAUTIONS



E / WD: FAILURE TITLE conditions	AURAL WARNING	MASTER LIGHT	SD PAGE CALLED	LOCAL WARNING	FLT PHASE INHIB
CIDS 1 + 2 FAULT	SINGLE CHIME	MASTER CAUTION	NIL	NIL	3,4,5 7,8
VHF 1(2) (3) EMITTING HF 1 (2) EMITTING Transmitter emitting more than 60 seconds.					
SATCOM FAULT (◁)	NIL	NIL			
ACARS FAULT (◁)					
HF 1(2) DATA FAULT (◁)					
VHF 3 DATA FAULT (◁)					
SATCOM DATA FAULT (◁)					

MEMO DISPLAY

This display shows “AUDIO 3 XFRD” in green, if the AUDIO SWITCHING selector is not on NORM.

If ACARS is installed, the display shows :

- “ACARS CALL” pulsing in green, if ACARS has received a message from the ground requesting a voice communication.
- “VHF 3 : VOICE” in green, if VHF3 is operating in voice mode and ACARS communication is interrupted.
- “ACARS MSG” pulsing in green, if ACARS has received a message from the ground.
- “ACARS STBY” in green, if ACARS communications between the aircraft and the ground are lost.

If the SATCOM is installed, the display shows “SATCOM ALERT” in green, when a message with a priority level below 4 is received from the ground.

If HF and ATSU are installed, the display shows “HF VOICE”, pulsing in green for 10 seconds and then steady, when all the HF radios are in voice mode.

BUS EQUIPMENT LIST

		NORM		EMER ELEC		
		AC	DC	AC ESS	DC ESS	HOT
RADIO COMMUNICATIONS	VHF1				X	
	VHF2		DC2			
	VHF3 ◁		DC1			
	HF1 ◁	AC1				
	HF2 ◁	AC2				
	RMP1				X	
	RMP2		DC2			
	RMP3 ◁		DC1			
	CAPT ACP				X	
	F/O ACP				X	
	THIRD ACP		DC1			
	FOURTH ACP ◁		DC1			
	SELCAL		DC1			
	FLT INTERPHONE				X	
	CAPT LOUDSPEAKER				X	
	F/O LOUDSPEAKER		DC2			
	EXT HORN					HOT 2
CABIN INTERCOM DATA SYS	CIDS1		GND/FLT		X	
	CIDS2		GND/FLT		X	
	DEU (A/B)		GND/FLT		X	
COCKPIT VOICE RECORDER	CVR CTL				SHED	
	CVR			SHED		
ACARS ◁		AC1				

BUS EQUIPMENT LIST

		NORM		EMER ELEC		
		AC	DC	AC ESS	DC ESS	HOT
RADIO COMMUNICATIONS	VHF1				X	
	VHF2		DC2			
	VHF3 ◁		DC1			
	HF1 ◁			SHED		
	HF2 ◁	AC2				
	RMP1				X	
	RMP2		DC2			
	RMP3 ◁		DC1			
	CAPT ACP				X	
	F/O ACP				X	
	THIRD ACP		DC1			
	FOURTH ACP ◁		DC1			
	SELCAL		DC1			
	FLT INTERPHONE				X	
	CAPT LOUDSPEAKER				X	
	F/O LOUDSPEAKER		DC2			
	EXT HORN					HOT 2
CABIN INTERCOMM DATA SYS	CIDS1		GND/FLT		X	
	CIDS2		GND/FLT		X	
	DEU (A/B)		GND/FLT		X	
COCKPIT VOICE RECORDER	CVR CTL				SHED	
	CVR			SHED		
ACARS ◁		AC1				